CHAPTER 07

Aldehydes and Ketones II: Enolates

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: fc6e37ba07712cb99bcaebc37cdc6bed6e4b6690067de5f60244403a789cc95a

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch07:exercise:alpha-map | alpha-map:stage-1, alpha-map:stage-2, alpha-map:stage-3

ch07:exercise:enolate-control | enolate-control:stage-1, enolate-control:stage-2, enolate-control:stage-3

ch07:exercise:aldol-map | aldol-map:stage-1, aldol-map:stage-2, aldol-map:stage-3

ch07:exercise:michael-site | michael-site:stage-1, michael-site:stage-2, michael-site:stage-3

ch07:lesson:alpha | alpha:step-1, alpha:step-2, alpha:step-3

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ch07:lesson:conjugate | conjugate:step-1, conjugate:step-2

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ch07:lesson:conjugate:cue | conjugate:cue

ch07:lesson:conjugate:trap | conjugate:trap

ch07:lesson:conjugate:app | conjugate:app

ch07:molecule:ketone | ketone

ch07:molecule:ethanal-enolate | ethanal-enolate

ch07:molecule:ethanal-enolate-carbon | ethanal-enolate-carbon

ch07:molecule:vinylalcohol | vinylalcohol

ch07:molecule:aldol | aldol

ch07:molecule:crotonaldehyde | crotonaldehyde

ch07:molecule:methylvinylketone | methylvinylketone

ch07:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 7.1 General Principles

scope-2: 7.2 Enolate Chemistry

scope-3: 7.3 Aldol Condensation

Objective: Identify alpha and beta positions

Objective: Distinguish tautomers from resonance and kinetic from thermodynamic control

Objective: Trace aldol and conjugate additions

Supplied Kaplan curriculum: Chapter 7; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter07b.xhtml

Enolate acidity cross-check: OpenStax 22.5 | https://openstax.org/books/organic-chemistry/pages/22-5-acidity-of-alpha-hydrogen-atoms-enolate-ion-formation

Aldehyde/ketone nucleophilic addition comparison: OpenStax 19.4, inspected | https://openstax.org/books/organic-chemistry/pages/19-4-nucleophilic-addition-reactions-of-aldehydes-and-ketones

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

alpha | Alpha carbons and acidic hydrogens

The conjugate base is resonance stabilized

alpha:rule | Core Rules

An alpha carbon is adjacent to a carbonyl carbon. Removing an alpha H gives an enolate stabilized by resonance and inductive withdrawal. For orientation, approximate teaching values are ethanal alpha-H pKa about 17 and acetone about 19; 1,3-dicarbonyl alpha H can be much more acidic. Values depend on substrate, solvent and scale; do not mix pKa tables from different media.

Visual card: α-Carbon Acidity (pKa 9–25), Enolate Resonance Hybrid & α-Halogenation / Haloform

Chemistry diagram: Resonance contributors of one enolate, not distinct species or tautomers. Only electron placement differs.

Ald/Ket α-H pKa ~19–20

β-Dicarbonyl α-H pKa ~9–11

Enolate: O^- (Major) ↔ C^- (Attacks C!)

α-Halogenation: Acid=1× vs Base=Haloform

Simple Ketone / Aldehyde | pKa ~19–20 (1 C=O resonance sink) | Needs LDA for 100% enolate (RO^- gives equilibrium)

Ester / Amide α-H | pKa ~25–30 (O/N donates into C=O) | Less acidic α-H than ketone/aldehyde

β-Diketone / β-Keto Ester | pKa ~9–11 (2 C=O resonance sinks!) | 100% deprotonated by mild EtO^- / HO^-

Why α-Hydrogens Are Exceptionally Acidic (pKa ~19–20 vs Alkane ~50) — — Deprotonating the α-carbon creates a resonance-stabilized enolate ion ($R–CO–CH_2^-$ ↔ $R–C(O^-)=CH_2$) that places negative charge on electronegative oxygen.

Active Methylene (β-Dicarbonyls, pKa ~9–11) — — A CH_2 flanked by TWO carbonyls delocalizes negative charge over BOTH oxygens, making it acidic enough for complete deprotonation by NaOEt or NaOH.

Ambident Enolate Reactivity — — Although oxygen holds more negative charge in the hybrid, carbon-electrophiles (R–X, C=O) react predominantly at the α-CARBON to preserve the strong C=O π bond!

alpha:cue | Memory Cues

Alpha = one bond from C=O. Carbonyl carbon is not alpha.

Visual card: Acid-Catalyzed vs. Base-Promoted α-Halogenation & Haloform Test

Are you halogenating a ketone in ACID or BASE?

Acid + Br_2 (CH_3COOH / H^+) — Stops Cleanly at MONO-α-Bromo Ketone! — Proceeds via the enol: adding the first inductive Br atom makes the carbonyl oxygen LESS basic, slowing down the second enolization.

Base + Excess Br_2/I_2 on Methyl Ketone ($R–CO–CH_3$) — Polyhalogenation → Haloform Cleavage ($CHX_3↓$ + $RCOO^-$) — Each inductive X makes the remaining α-protons MORE acidic! All 3 α-H atoms are replaced ($–CX_3$), and HO^- cleaves it into carboxylate + yellow CHI_3 precipitate.

Chiral α-Carbon in Acid or Base — Equilibrates to Racemic Mixture via Planar Enol/Enolate! — Only stereocenters AT the α-carbon with at least one α-H can racemize; β/γ stereocenters never racemize via enolization.

alpha:trap | Common Mistakes / MCAT Traps

Only a site bearing H can be deprotonated there. Stabilization of an enolate explains alpha acidity; do not treat every C–H bond in a carbonyl molecule as equally acidic. Among comparable simple examples, aldehyde alpha H is often more acidic than ketone alpha H. Additional alkyl donation can destabilize the conjugate base; substituents and conditions can change the trend. Aldehydes also often undergo nucleophilic addition more readily than comparable ketones because they are less

crowded and less electron-rich.

Visual card: Using Base + Br_2 for Monobromination or Racemizing Non-α Stereocenters

Common Trap — — Using NaOH + 1 eq Br_2 expecting clean monobromination, or claiming a β-chiral carbon racemizes in base.

Correct Reasoning — — Base-promoted halogenation accelerates after the first halogen is added (giving polyhalogenation); only α-stereocenters bearing an α-H can racemize via planar enols/enolates.

alpha:app | MCAT Applications

Enolate-like intermediates allow enzymes to make or break carbon–carbon bonds adjacent to carbonyls.

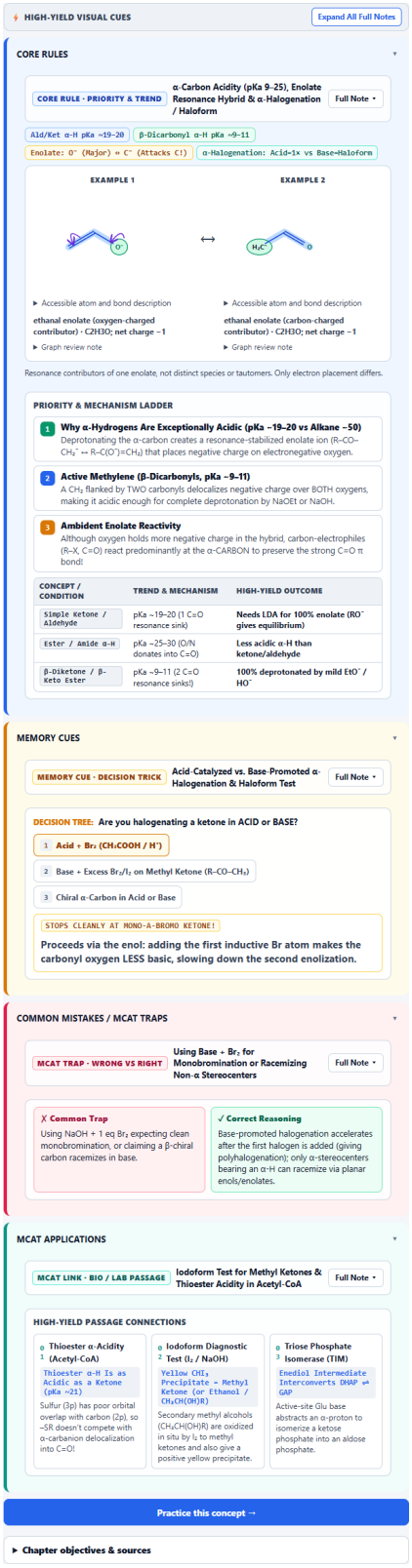
Visual card: Iodoform Test for Methyl Ketones & Thioester Acidity in Acetyl-CoA

Thioester α-Acidity (Acetyl-CoA) — Thioester α-H Is as Acidic as a Ketone (pKa ~21) — Sulfur (3p) has poor orbital overlap with carbon (2p), so –SR doesn't compete with α-carbanion delocalization into C=O!

Iodoform Diagnostic Test (I_2 / NaOH) — Yellow CHI_3 Precipitate = Methyl Ketone (or Ethanol / $CH_3CH(OH)R$) — Secondary methyl alcohols ($CH_3CH(OH)R$) are oxidized in situ by I_2 to methyl ketones and also give a positive yellow precipitate.

Triose Phosphate Isomerase (TIM) — Enediol Intermediate Interconverts DHAP ⇌ GAP — Active-site Glu base abstracts an α-proton to isomerize a ketose phosphate into an aldose phosphate.

Screenshot: alpha-visuals



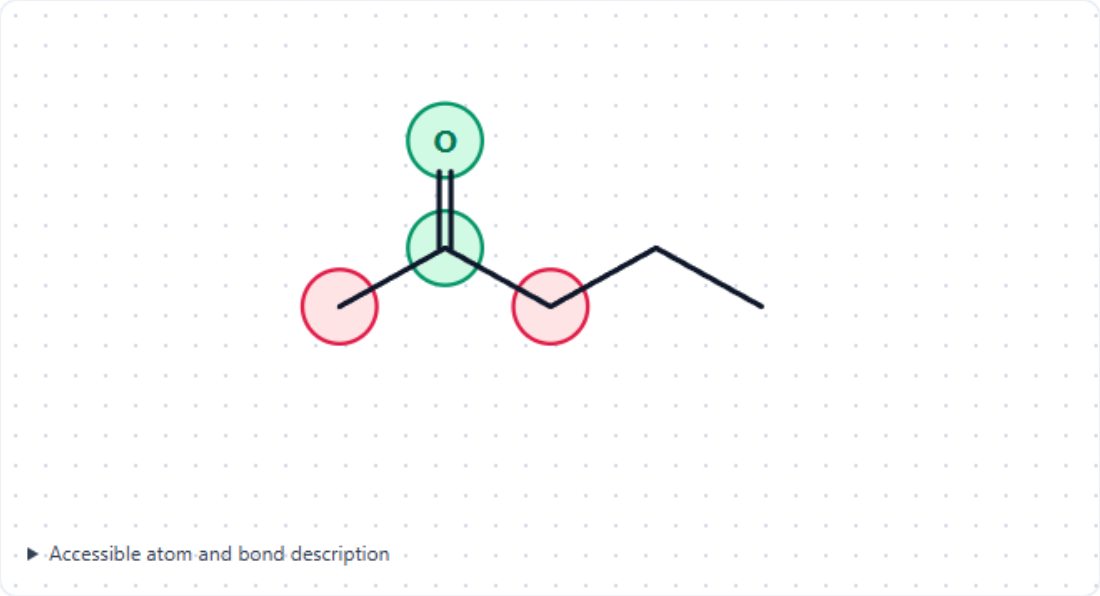
alpha:step-1 | 1. Locate alpha sites

Pentan-2-one has alpha carbons on both sides of its carbonyl: the terminal methyl and the neighboring methylene.

Displayed molecule ID: ketone

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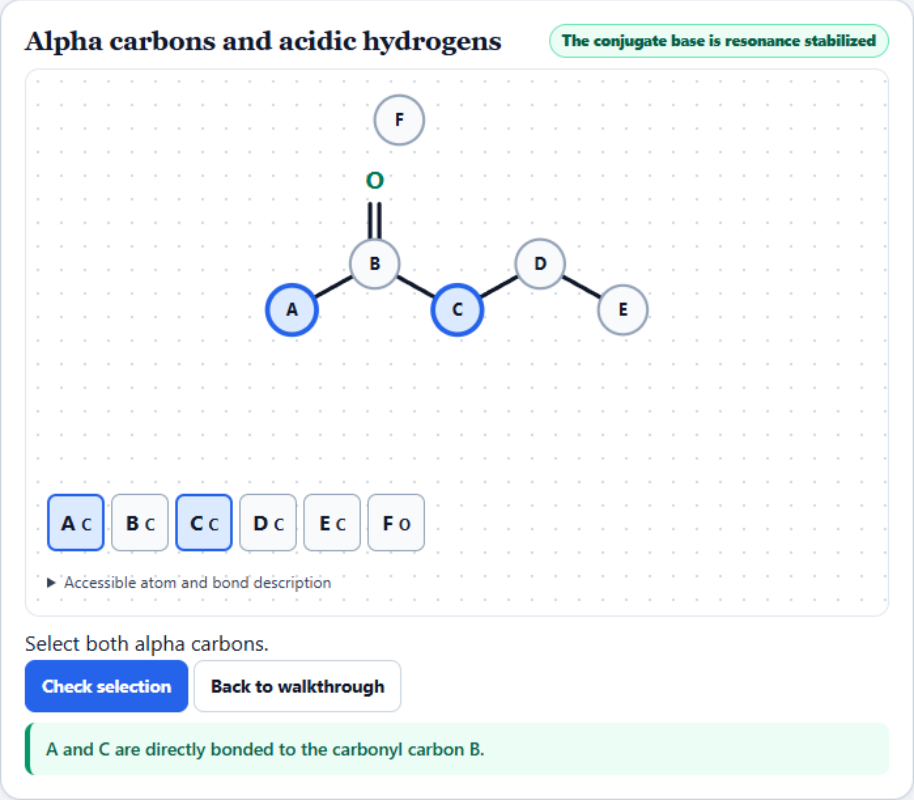
Screenshot: alpha-step-1



Embedded challenge: Select both alpha carbons.

{"prompt": "Select both alpha carbons.", "answer": [0, 2], "explanation": "A and C are directly bonded to the carbonyl carbon B."}

Screenshot: alpha-step-1-challenge



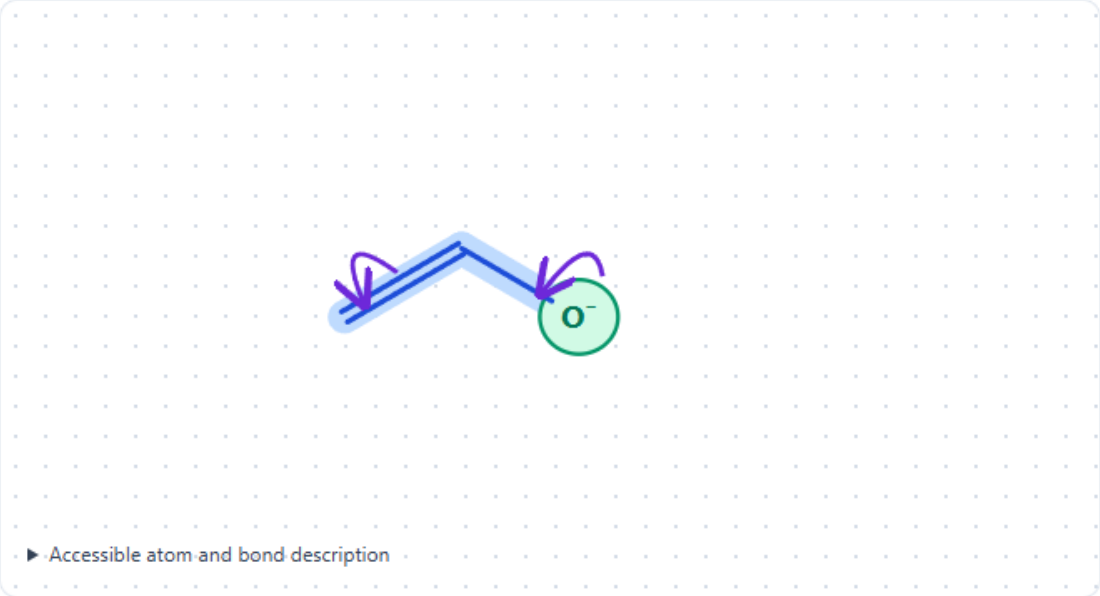
alpha:step-2 | 2. Delocalize charge

This oxygen-charged contributor and the carbon-charged contributor in the next step describe one delocalized ethanal enolate. Purple arrows redistribute electrons, not atoms; no proton moves in this resonance operation.

Displayed molecule ID: ethanal-enolate

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Screenshot: alpha-step-2



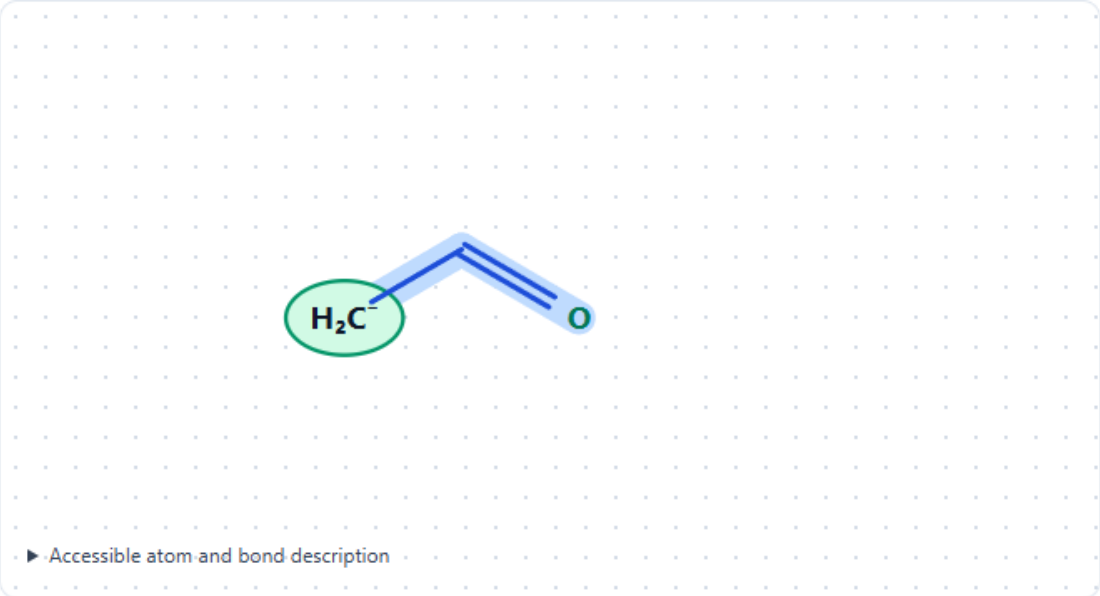
alpha:step-3 | 3. Compare the carbon contributor

The same atom skeleton now has a carbonyl and a lone pair/negative formal charge on the alpha carbon, shown as H2C−. Both contributors have C2H3O composition and net −1 charge. They are not separate species interconverting in time; the actual ion is delocalized.

Displayed molecule ID: ethanal-enolate-carbon

{"highlights": {"prin": [0], "path": [0, 1, 2]}, "revealName": true}

Screenshot: alpha-step-3



tautomers | Tautomers and enolate selectivity

Kinetic control vs. thermodynamic control

tautomers:rule | Core Rules

Keto and enol forms are constitutional isomers that differ in H placement and a double bond; interconversion requires proton transfer. An enolate is the deprotonated species, not another neutral tautomer. Bulky strong base at low temperature under irreversible conditions often favors the less substituted kinetic enolate. Reversible conditions often favor the more substituted thermodynamic enolate.

Visual card: Keto–Enol Tautomerism vs. Resonance & Kinetic vs. Thermodynamic Enolate Control

Chemistry diagram: Illustrative enol structure. Tautomerization changes H and pi-bond placement; resonance does not move atoms.

Tautomers = Constitutional Isomers (Move H + π)

Resonance = Move e^- Only

Kinetic = LDA, −78°C (Less Hindered)

Thermo = NaOEt, 25°C (More Subst)

Keto vs Enol Tautomers | Constitutional isomers in equilibrium (⇌) | Keto heavily favored (C=O + C–H stronger than C=C + O–H)

Kinetic Enolate | LDA, THF, −78 °C (Irreversible, Bulky) | Deprotonates LESS hindered α-C (fastest rate, lower Ea)

Thermodynamic Enolate | NaOR / KH, 25 °C (Reversible, Equilibrating) | Forms MORE substituted C=C double bond (lowest energy)

Tautomers Are Constitutional Isomers (NOT Resonance Contributors!) — — Tautomerism moves a PROTON (α-H ↔ O–H) and shifts a π bond (⇌ equilibrium arrow); resonance moves ONLY electrons with all nuclei fixed (↔ double-headed arrow).

Why Keto Usually Dominates Over Enol (>99%) — — The sum of bond energies for C=O + C–H is ~45–60 kJ/mol stronger than C=C + O–H. Exceptions: phenol (aromaticity!) and 1,3-diketones (conjugated + intramolecular H-bond!).

Kinetic vs. Thermodynamic Enolate Regioselectivity — — Unsymmetrical ketones (2-methylcyclohexanone) give the LESS substituted enolate with bulky LDA at −78 °C (kinetic) and the MORE substituted enolate with NaOEt at 25 °C (thermodynamic).

tautomers:cue | Memory Cues

Tautomers move H and pi bonds. Resonance moves only electrons.

Visual card: Kinetic vs. Thermodynamic Enolate Reaction Conditions Selector

Look at the base, solvent, and temperature over the reaction arrow:

LDA (Lithium Diisopropylamide), THF, −78 °C — KINETIC Enolate (Less Substituted α-Carbon) — Hindered strong base plucks the most sterically accessible α-H irreversibly at −78 °C before equilibration can occur.

NaOEt / EtOH or KH, Room Temp / Heat (25 °C) — THERMODYNAMIC Enolate (More Substituted α-Carbon) — Reversible proton transfer allows equilibration to the more alkyl-substituted, lower-energy internal C=C enolate.

Phenol or 2,4-Pentanedione (β-Diketone) — Enol Tautomer Is Unusually Favored! — Phenol is ~100% enol to preserve aromaticity; β-diketones favor the enol via conjugation + a 6-membered intramolecular H-bond ring.

tautomers:trap | Common Mistakes / MCAT Traps

Kinetic/thermodynamic predictions require the substrate and conditions. More substituted is not a universal stability rule for every enolate. Simple aldehydes and ketones usually favor the keto form, but conjugation and aromaticity can change

that. Enolization makes an alpha center planar and can erase its configuration. Reprotonation from either face can racemize a single alpha stereocenter in an otherwise achiral environment; with other stereocenters it can cause epimerization. Rate and extent depend on conditions, so rapid complete racemization is not automatic.

Visual card: Calling Keto and Enol "Resonance Structures" or Using ↔ Instead of ⇌

Common Trap — — Saying "acetone is stabilized by keto-enol resonance" or drawing a double-headed resonance arrow ↔ between a ketone and its enol.

Correct Reasoning — — Keto and enol forms have different H atom connectivity (constitutional isomers that can be observed separately in NMR); Enolate anions are resonance hybrids!

tautomers:app | MCAT Applications

Selective enolate generation lets a synthesis control which alpha carbon forms a new bond.

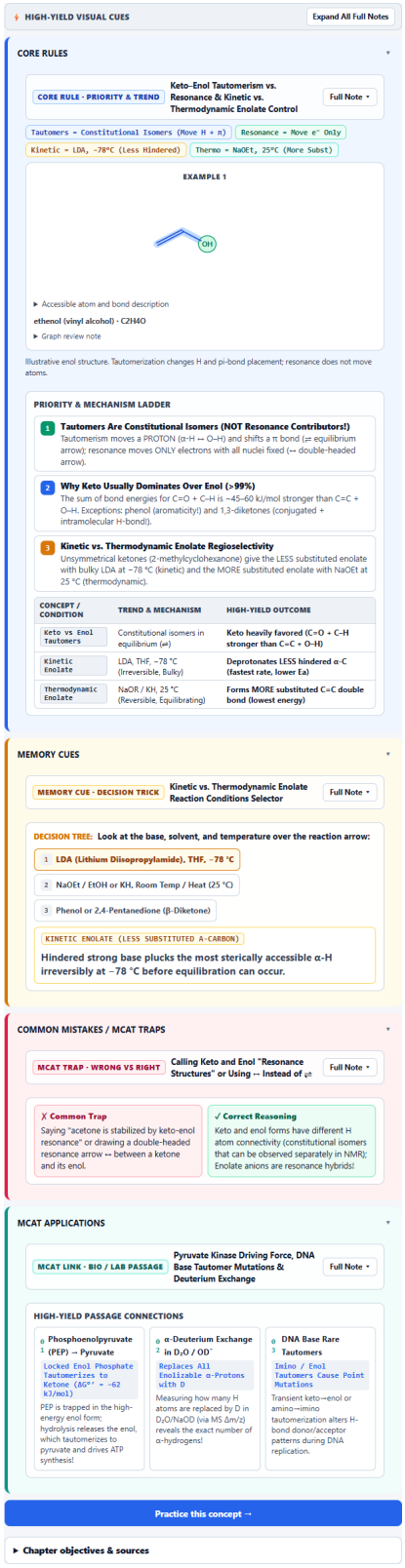
Visual card: Pyruvate Kinase Driving Force, DNA Base Tautomer Mutations & Deuterium Exchange

Phosphoenolpyruvate (PEP) → Pyruvate — Locked Enol Phosphate Tautomerizes to Ketone ($\Delta G^{\circ\prime} = -62$ kJ/mol) — PEP is trapped in the high-energy enol form; hydrolysis releases the enol, which tautomerizes to pyruvate and drives ATP synthesis!

α-Deuterium Exchange in D_2O / OD^- — Replaces All Enolizable α-Protons with D — Measuring how many H atoms are replaced by D in D_2O/NaOD (via MS Δm/z) reveals the exact number of α-hydrogens!

DNA Base Rare Tautomers — Imino / Enol Tautomers Cause Point Mutations — Transient keto→enol or amino→imino tautomerization alters H-bond donor/acceptor patterns during DNA replication.

Screenshot: tautomers-visuals



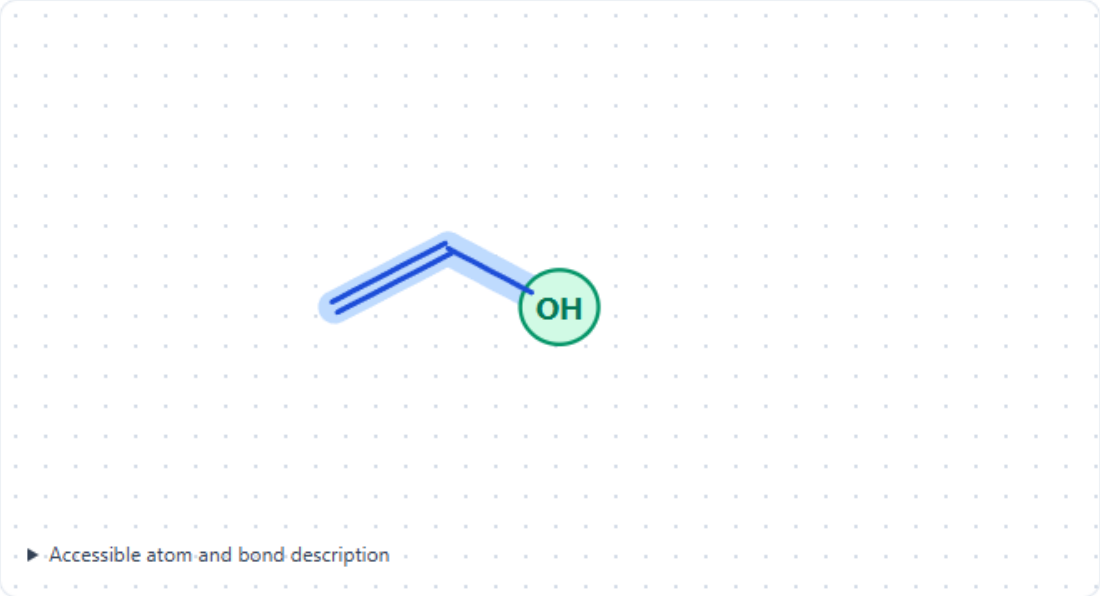
tautomers:step-1 | 1. Keto vs. enol

Ethanal and ethenol have the same formula but different H and bond placement. This enol has OH directly attached to an alkene carbon.

Displayed molecule ID: vinylalcohol

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Screenshot: tautomers-step-1



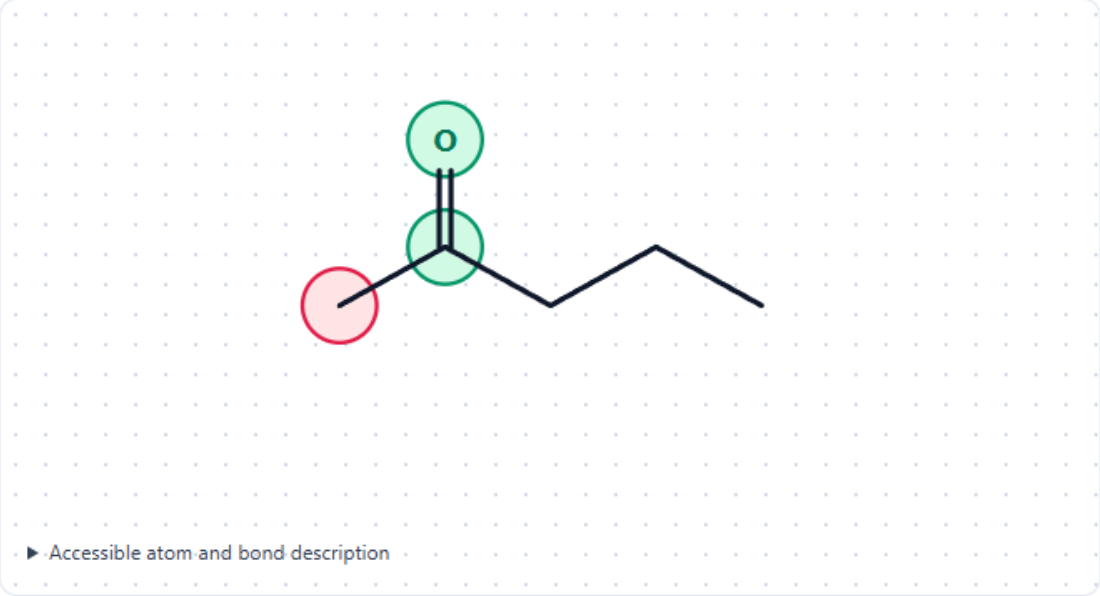
tautomers:step-2 | 2. Kinetic site

For pentan-2-one, bulky LDA under typical low-temperature kinetic conditions often removes an H from the less hindered methyl side.

Displayed molecule ID: ketone

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Screenshot: tautomers-step-2



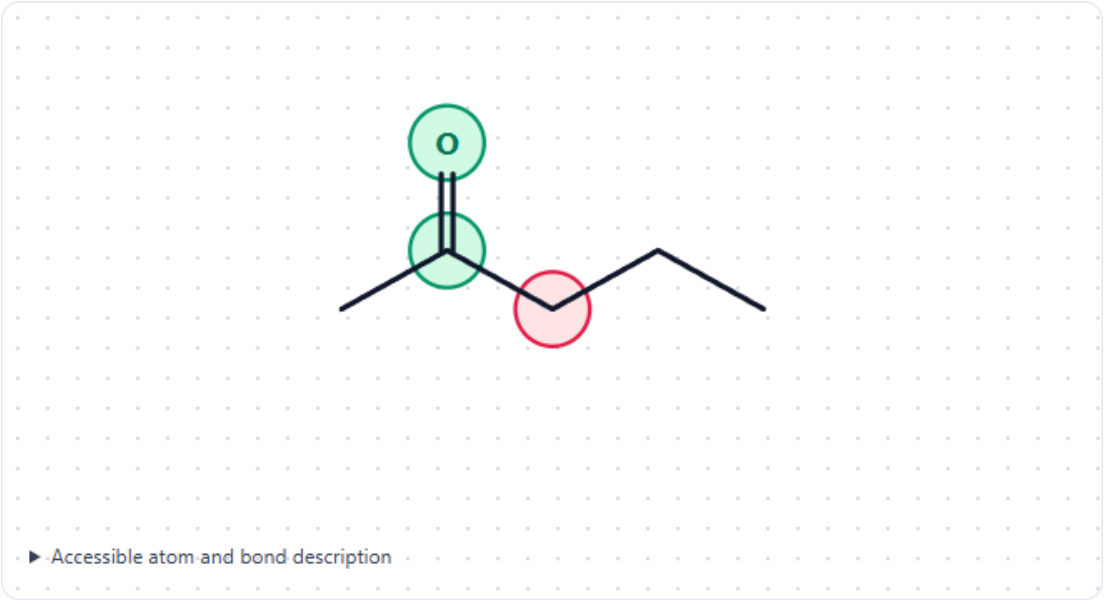
tautomers:step-3 | 3. Thermodynamic site

Under suitable reversible conditions, the more substituted enolate from the methylene side is often favored. State the conditions before predicting selectivity.

Displayed molecule ID: ketone

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Screenshot: tautomers-step-3



aldol | Aldol addition, condensation, and retro-aldol

New bond: donor alpha C to acceptor carbonyl C

aldol:rule | Core Rules

An enolate adds to another carbonyl and is protonated to give a beta-hydroxy aldehyde or ketone: aldol addition. Subsequent dehydration gives an alpha,beta-unsaturated carbonyl: aldol condensation. Retro-aldol cleavage breaks the alpha–beta C–C bond of a beta-hydroxy carbonyl.

Visual card: Aldol Addition (β-Hydroxy Carbonyl), Dehydration Condensation (α,β-Enone) & Retro-Aldol

Chemistry diagram: Suitable dehydration removes water from a beta-hydroxy carbonyl to give an unsaturated carbonyl; schematic net pathway.

Aldol Addition → β-Hydroxy Carbonyl

Condensation (+Δ, -H_2O) → α,β-Unsaturated C=O

Retro-Aldol = Cleave α–β Bond!

Aldol Addition (Cold/Mild) | Enolate + Ald/Ket → β-Hydroxy Carbonyl | Forms new Cα–Cβ σ bond (100% atom economy)

Aldol Condensation (Heat Δ) | Eliminates H_2O → α,β-Unsaturated C=O | Driven by conjugated C=C–C=O π system

Retro-Aldol Reaction | Base/Enzyme cleaves Cα–Cβ bond | Turns β-hydroxy/C=C–C=O back into 2 carbonyls

Step 1: Nucleophilic Addition (Cα Attacks Partner C=O) — — An α-enolate (or enol in acid) attacks an electrophilic carbonyl carbon to form a new Cα–Cβ bond and a β-hydroxy aldehyde/ketone ("ald-ol").

Step 2: Dehydration to α,β-Unsaturated Carbonyl (Heat Δ) — — Warming in base (E1cb via enolate expulsion of HO^-) or acid eliminates H_2O to form a conjugated C=C–C=O system.

Crossed Aldol & Intramolecular Aldol Ring Size — — Clean crossed aldols use one partner with NO α-H (benzaldehyde, formaldehyde) OR pre-form a directed kinetic enolate with LDA at −78 °C. Intramolecular aldols favor 5- and 6-membered rings.

aldol:cue | Memory Cues

Addition adds the partner. Condensation also loses water. The new OH is beta to the retained carbonyl.

Visual card: 3-Step Retro-Aldol Disconnection Trick for Any Exam Structure

How do you find the starting carbonyls of an aldol product?

1. Label the Carbonyl Reference — C=O Is Reference → Adjacent = α → Next = β — In a valid aldol product, the β-carbon MUST bear either an –OH group (addition) or a C=C double bond to α (condensation).

2. Break the Cα–Cβ Bond — Cut Straight Through the α–β Bond — The α-carbon piece was the NUCLEOPHILIC enolate; the β-carbon piece was the ELECTROPHILIC carbonyl partner.

3. Restore the Original Functional Groups — α-Carbon Gets H (or H_2); β-Carbon Becomes C=O! — Convert the β–OH (or β=α alkene) back into a C=O double bond to reveal the two starting aldehydes/ketones!

aldol:trap | Common Mistakes / MCAT Traps

Aldol products need not contain an aldehyde: ketone donors/acceptors can give beta-hydroxy ketones. Mixed aldols may give mixtures unless donor/acceptor roles are controlled.

Visual card: Drawing a γ-Hydroxy Carbonyl or Forgetting Why HO^- Can Leave in E1cb Dehydration

Common Trap — — Connecting the carbonyl carbon or β-carbon instead of the α-carbon, or claiming HO^- can never leave under any circumstances.

Correct Reasoning — — Aldol ALWAYS forms a 1,3-functionalized (β-hydroxy or α,β-unsaturated) product; in E1cb dehydration, a stabilized α-enolate pushes out HO^- to gain C=C–C=O conjugation!

aldol:app | MCAT Applications

Aldolase cleavage of carbon skeletons uses this bond logic in metabolism, with enzyme-specific catalysis.

Visual card: Glycolysis Aldolase (FBP ⇌ DHAP + GAP) & Citrate Synthase / PLP

Fructose-1,6-Bisphosphate Aldolase — Reversible Retro-Aldol (6C ⇌ 3C + 3C) — Cleaves the C3–C4 (α–β) bond of fructose-1,6-bisphosphate into DHAP (enamine/enolate donor) and GAP (aldehyde acceptor).

Class I vs. Class II Aldolases — Schiff Base Enamine (Lys) vs. Zn^{2+} Enolate — Human Class I aldolase forms a protonated Schiff base iminium with Lys to act as an electron sink; bacterial Class II uses a Zn^{2+} cofactor.

Intramolecular Robinson Annulation — Michael Addition + Intramolecular Aldol — Fuses a new 6-membered cyclohexenone ring onto an existing ketone scaffold in steroid synthesis.

Screenshot: aldol-visuals



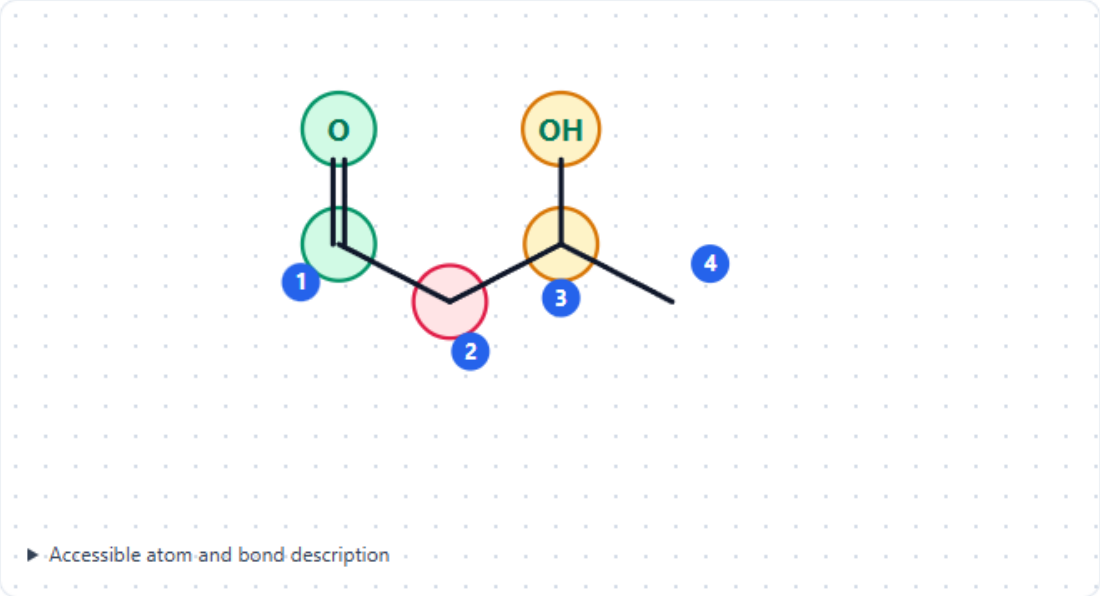
aldol:step-1 | 1. Addition product

Two ethanal units can form 3-hydroxybutanal. The donor retains its carbonyl; the acceptor carbonyl becomes the beta-OH carbon. The newly formed bond joins C2 and C3.

Displayed molecule ID: aldol

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Screenshot: aldol-step-1



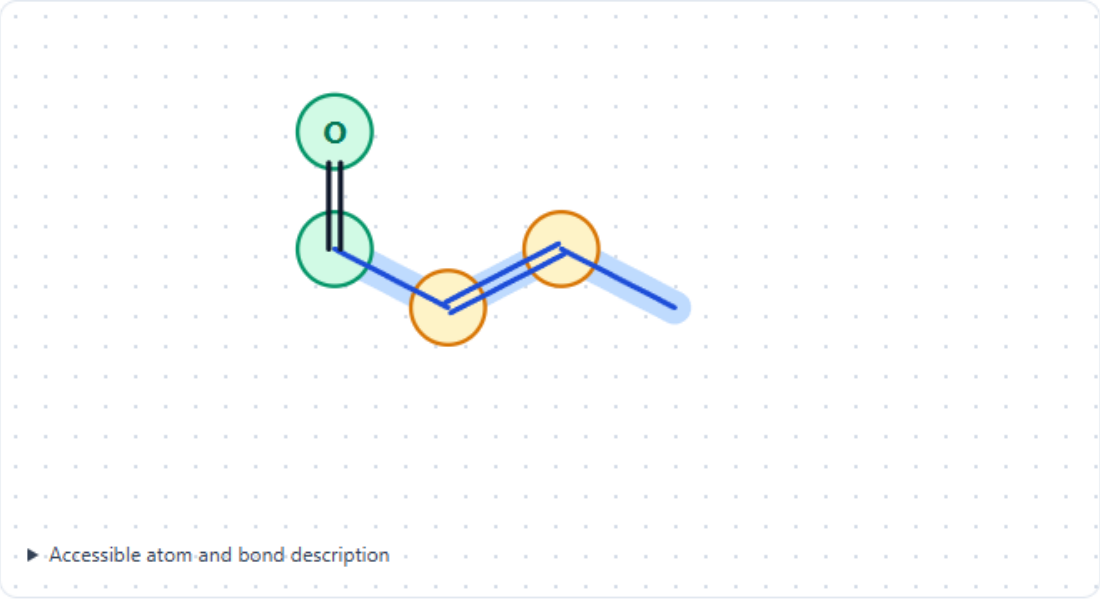
aldol:step-2 | 2. Dehydrate

Loss of water from the alpha/beta region gives a conjugated C=C–C=O system. The diagram specifies connectivity only, not an E/Z assignment.

Displayed molecule ID: crotonaldehyde

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Screenshot: aldol-step-2



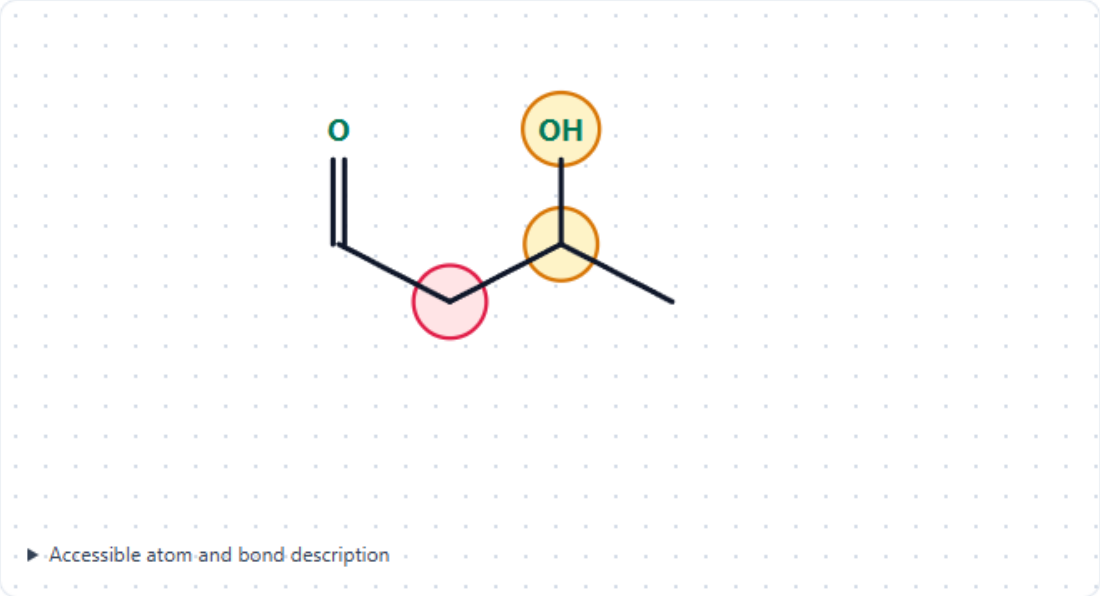
aldol:step-3 | 3. Reverse the carbon–carbon step

In a retro-aldol, cleave the bond between alpha and beta carbons of the beta-hydroxy carbonyl; regenerate carbonyl-containing fragments.

Displayed molecule ID: aldol

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Screenshot: aldol-step-3



conjugate | Conjugate addition and enamine nucleophiles

An enone has more than one electrophilic site

conjugate:rule | Core Rules

An alpha,beta-unsaturated carbonyl can undergo direct 1,2-addition at carbonyl carbon or conjugate 1,4-addition at beta carbon. Enolates and enamines can serve as carbon nucleophiles; their selectivity depends on reagent and conditions.

Visual card: 1,2-Direct vs. 1,4-Conjugate (Michael) Addition & The Stork Enamine Synthesis

Chemistry diagram: Carbonyl-carbon and beta-carbon focus on the same enone: possible 1,2 versus 1,4 sites, not reaction products.

1,2-Addition = Carbonyl C (Hard/Fast)

1,4-Michael = β-Carbon (Soft/Reversible)

Gilman (R_2CuLi) → 1,4

RLi / RMgBr → 1,2

1,2-Direct Addition | Attacks C=O carbon (Kinetic control) | Organolithium (RLi), Grignard (RMgBr), $LiAlH_4$

1,4-Conjugate (Michael) | Attacks β-carbon (Thermodynamic) | Gilman (R_2CuLi), Enolates, Enamines, RSH, R_2NH, CN^-

Stork Enamine Reaction | Enamine + Michael acceptor → 1,5-Dicarbonyl | Mild, neutral alternative to strong-base enolates

Two Electrophilic Sites in α,β-Unsaturated Carbonyls — — Resonance (C=C–C=O ↔ $^+$Cβ–Cα=C–O^-) places δ+ charge on BOTH the carbonyl carbon (1,2-site) and the alkene β-carbon (1,4-site).

1,2- vs. 1,4-Regioselectivity (Hard/Irreversible vs. Soft/Reversible) — — Fierce, irreversible nucleophiles (RLi, RMgBr, $LiAlH_4$) attack the carbonyl C (1,2); softer or stabilized nucleophiles (R_2CuLi, β-dicarbonyl enolates, enamines, thiols, amines) add to the β-carbon (1,4) to preserve the strong C=O bond!

Michael Addition + Aldol = Robinson Annulation — — 1,4-Michael addition of an enolate to methyl vinyl ketone forms a 1,5-diketone, which cyclizes via intramolecular aldol condensation into a 6-membered cyclohexenone ring.

conjugate:cue | Memory Cues

Michael addition: nucleophile bonds to beta carbon, then protonation/tautomerization restores the carbonyl.

Visual card: 1,2-Direct vs. 1,4-Conjugate Nucleophile Classifier

Which nucleophile is reacting with cyclohex-2-en-1-one?

Organolithium (CH_3Li) or Grignard (RMgBr) — 1,2-Addition to Carbonyl Carbon → Allylic Alcohol — Highly basic, localized carbanions attack the highest δ+ center (C=O) irreversibly under kinetic control.

Gilman Reagent (R_2CuLi), Thiol (RSH), or Amine — 1,4-Conjugate Addition to β-Carbon → Saturated Ketone — Soft nucleophiles add at the β-carbon to form a resonance-stabilized enolate, which tautomerizes to keep the strong C=O π bond!

Stabilized Enolate (β-Diketone) or Stork Enamine — Michael 1,4-Addition → 1,5-Dicarbonyl Product! — Wait: Aldol gives 1,3-functionalized products; Michael conjugate addition ALWAYS gives a 1,5-dicarbonyl spacing!

conjugate:trap | Common Mistakes / MCAT Traps

Do not label every nucleophile addition to an enone “Michael.” Direct carbonyl attack is 1,2-addition. Enamines are neutral nitrogen analogues used for alpha functionalization, then hydrolyzed back to carbonyls. A suitable imine with an alpha H can tautomerize to an enamine with N–H by proton and double-bond relocation. Relative tautomer stability depends on substitution and conjugation. A tertiary enamine without N–H does not have an ordinary neutral imine tautomer; protonation instead gives an iminium ion.

Visual card: Thinking "1,4-Addition" Leaves the Nucleophile on Oxygen

Common Trap — — Attaching the nucleophile to Cβ and leaving a permanent substituent on O (or forgetting tautomerization back to C=O).

Correct Reasoning — — In 1,4-conjugate addition, the nucleophile bonds to Cβ and a proton initially lands on O (forming a 1,4-enol), which immediately tautomerizes to put H on Cα and restore C=O (net 3,4-addition across C=C)!

conjugate:app | MCAT Applications

Conjugate acceptors can react with biological nucleophiles; the site of reaction matters for covalent inhibitor design.

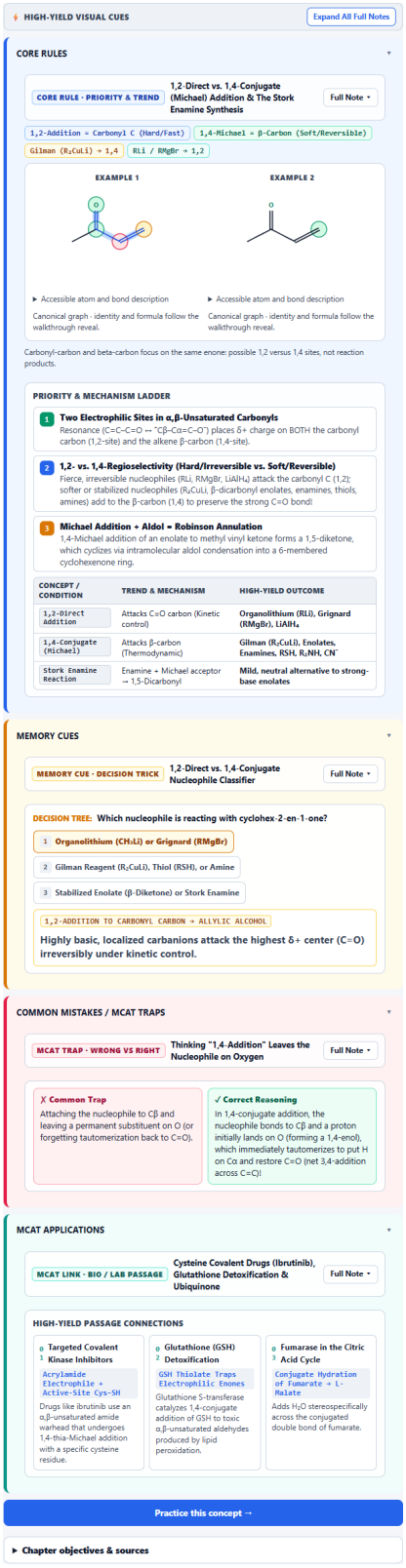
Visual card: Cysteine Covalent Drugs (Ibrutinib), Glutathione Detoxification & Ubiquinone

Targeted Covalent Kinase Inhibitors — Acrylamide Electrophile + Active-Site Cys–SH — Drugs like ibrutinib use an α,β-unsaturated amide warhead that undergoes 1,4-thia-Michael addition with a specific cysteine residue.

Glutathione (GSH) Detoxification — GSH Thiolate Traps Electrophilic Enones — Glutathione S-transferase catalyzes 1,4-conjugate addition of GSH to toxic α,β-unsaturated aldehydes produced by lipid peroxidation.

Fumarase in the Citric Acid Cycle — Conjugate Hydration of Fumarate → L-Malate — Adds H_2O stereospecifically across the conjugated double bond of fumarate.

Screenshot: conjugate-visuals



conjugate:step-1 | 1. Locate the conjugated system

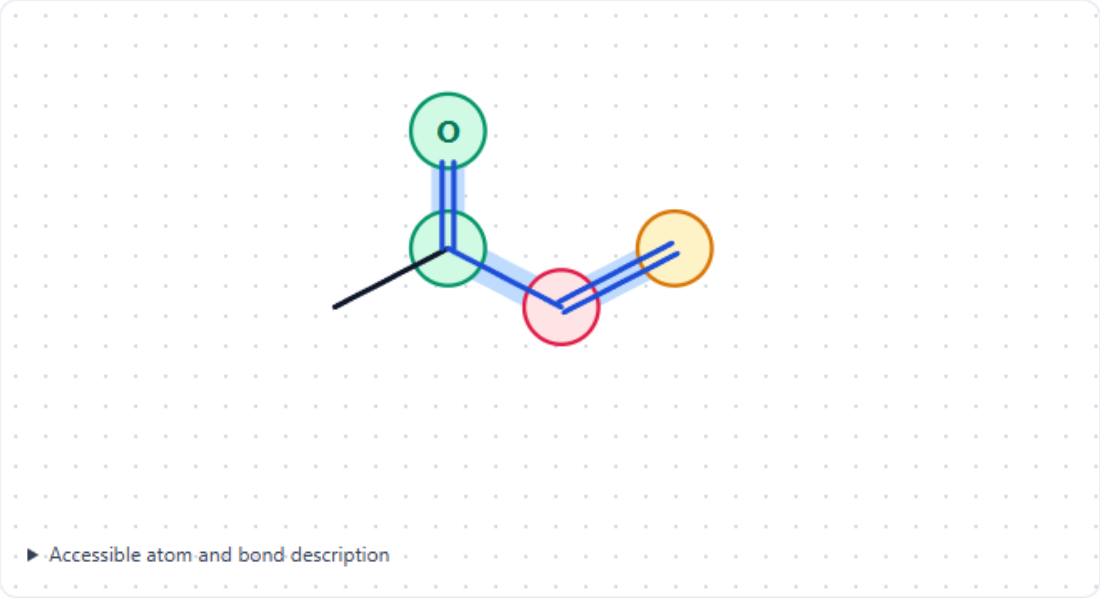
The carbonyl group is green, its adjacent alpha carbon is red, and the terminal beta carbon is orange. Locate them by their bonds as well as color: alpha is one C–C bond from the carbonyl carbon, beta is two. Aligned p orbitals allow electron

delocalization across the alternating bonds.

Displayed molecule ID: methylvinylketone

{"highlights": {"path": [4, 1, 2, 3], "prin": [1, 4], "alpha": [2], "sub": [3]}, "revealName": true}

Screenshot: conjugate-step-1



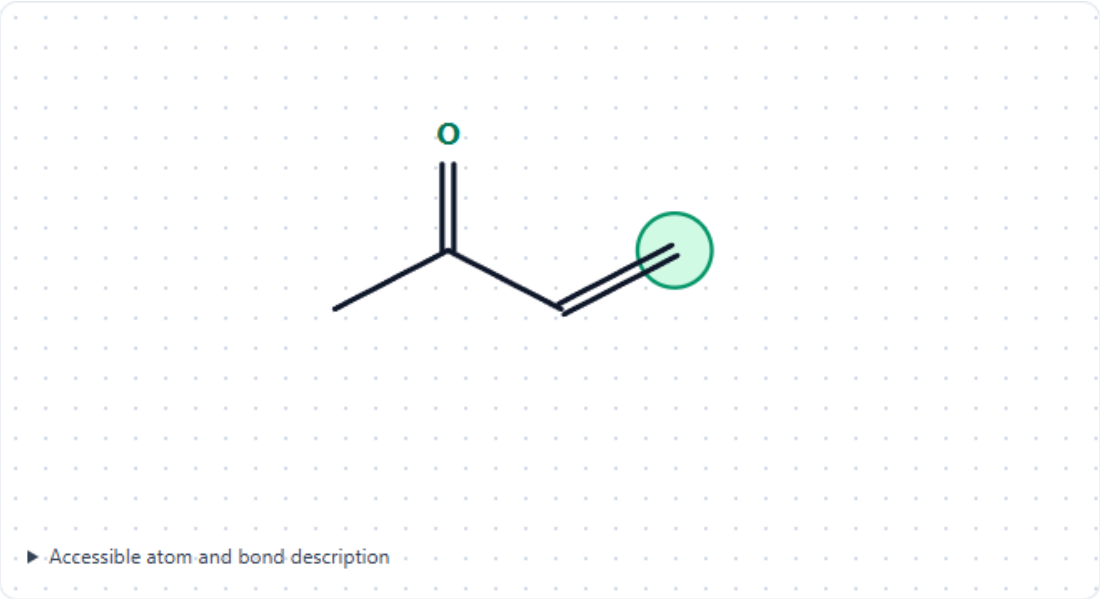
conjugate:step-2 | 2. Choose the beta site

The beta carbon, now the green focus, is the terminal alkene carbon two C–C bonds from the carbonyl carbon. A Michael-type addition makes a new bond there; protonation of the enolate-like intermediate restores the carbonyl. Green marks the current focus rather than one fixed atom type.

Displayed molecule ID: methylvinylketone

{"highlights": {"prin": [3]}, "revealName": false}

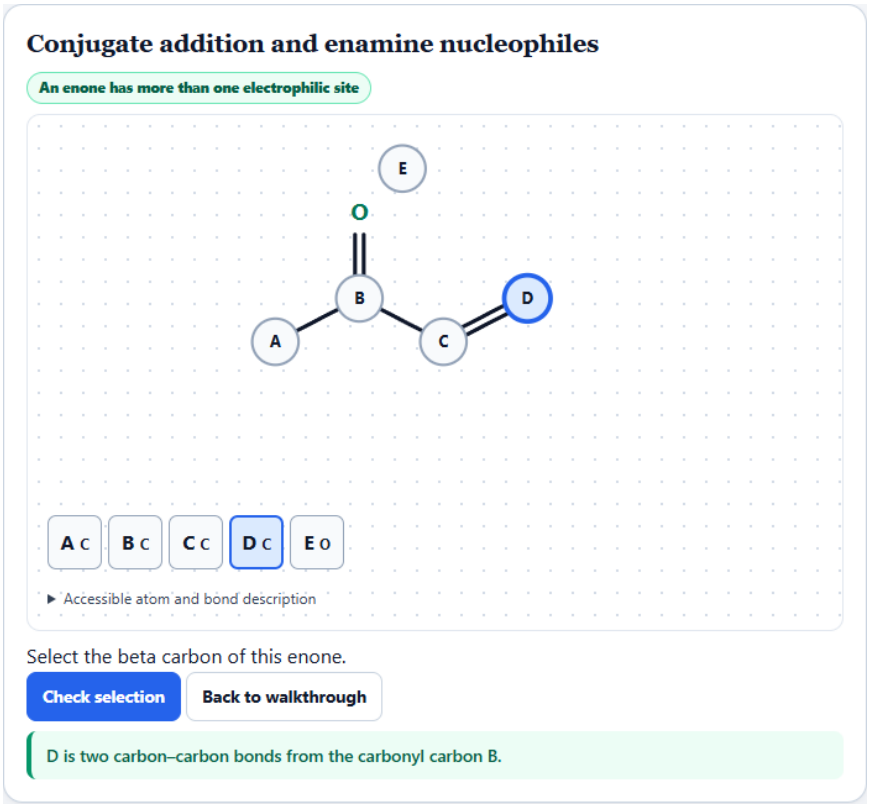
Screenshot: conjugate-step-2



Embedded challenge: Select the beta carbon of this enone.

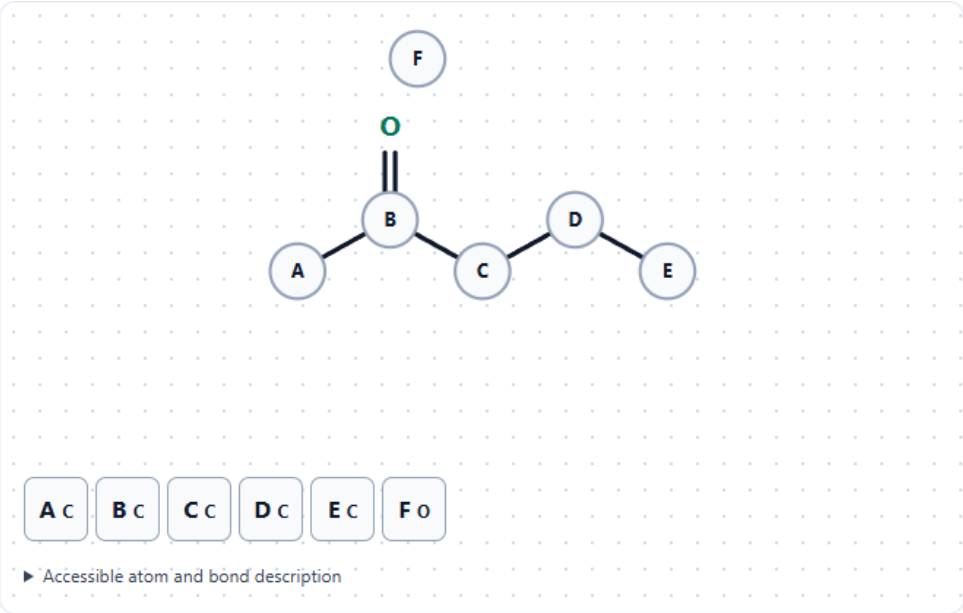
{"prompt": "Select the beta carbon of this enone.", "answer": [3], "explanation": "D is two carbon–carbon bonds from the carbonyl carbon B."}

Screenshot: conjugate-step-2-challenge



alpha-map | Identify enolate-forming sites

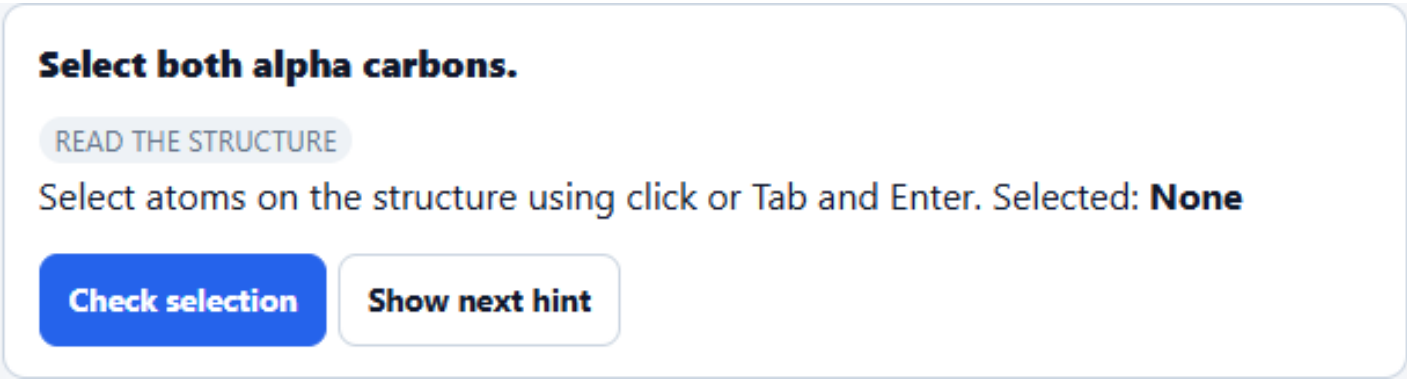
Screenshot: alpha-map-stage-1-molecule



alpha-map:stage-1 | Select both alpha carbons.

Answer type: atoms

Screenshot: alpha-map-stage-1-question



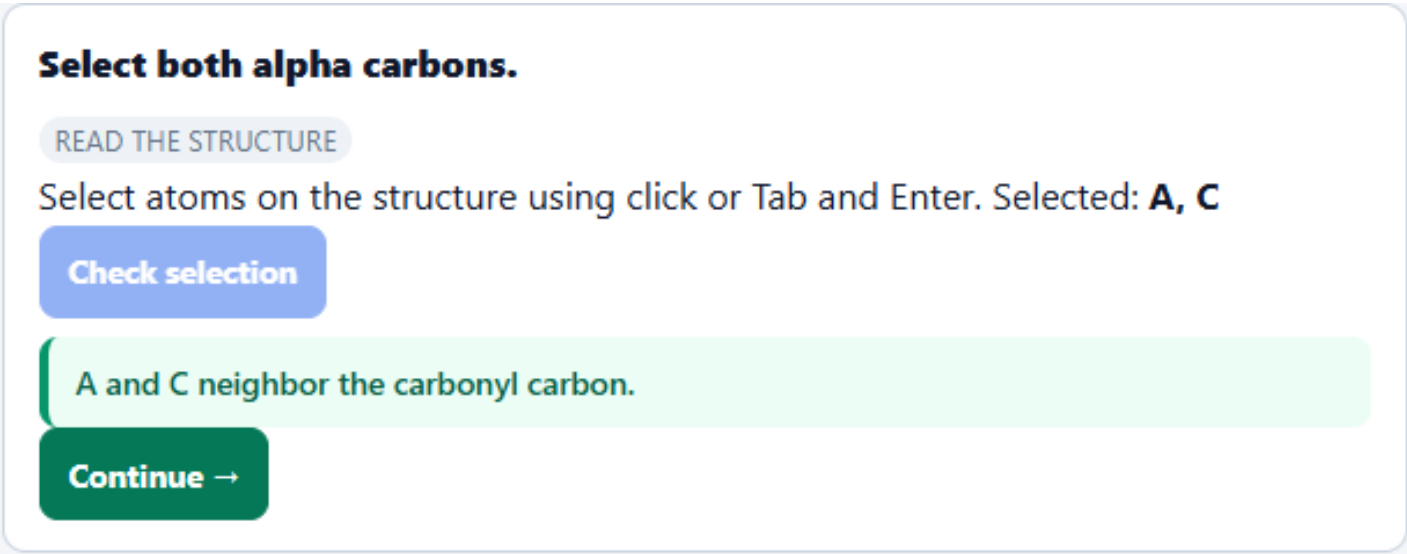
{"authorAnswer": [0, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"alpha": [0, 2]}}

Hint 1: Locate C=O first.

Hint 2: Choose carbon neighbors of that carbon.

Explanation: A and C neighbor the carbonyl carbon.

Screenshot: alpha-map-stage-1-feedback



alpha-map:stage-2 | Why are their hydrogens relatively acidic?

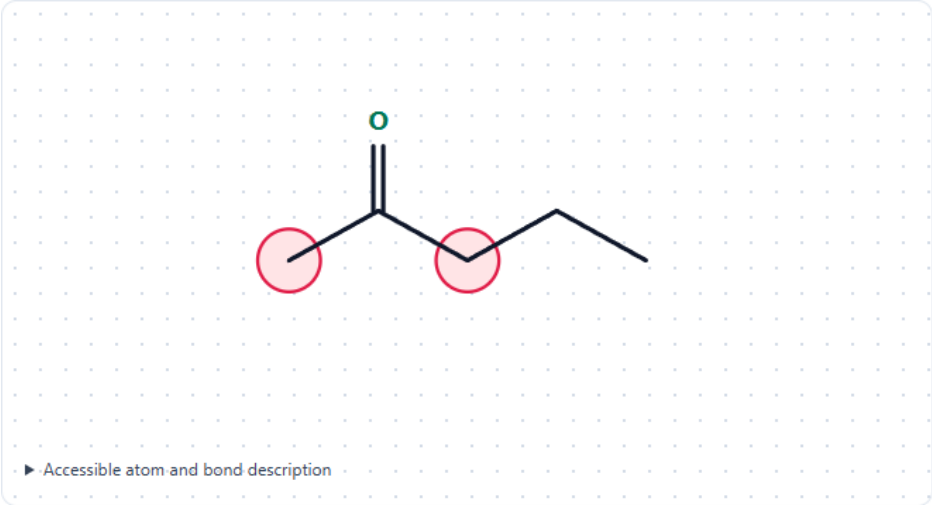
Answer type: choice

Option 0 (A): The conjugate base is resonance stabilized

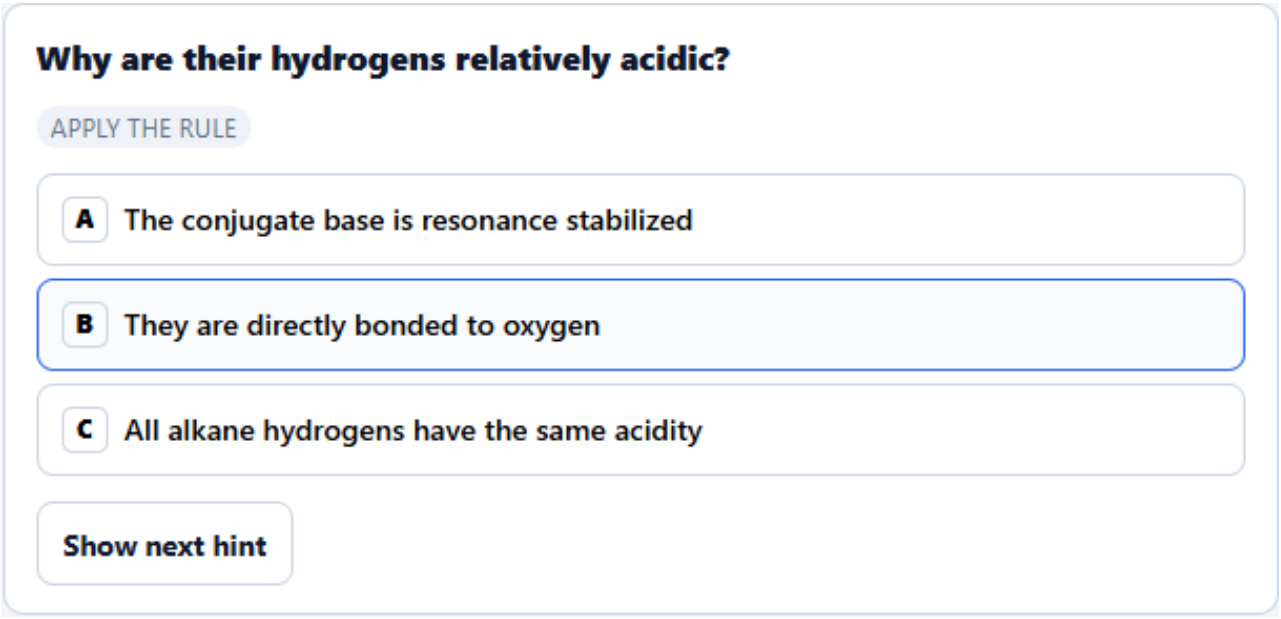
Option 1 (B): They are directly bonded to oxygen

Option 2 (C): All alkane hydrogens have the same acidity

Screenshot: alpha-map-stage-2-molecule



Screenshot: alpha-map-stage-2-question



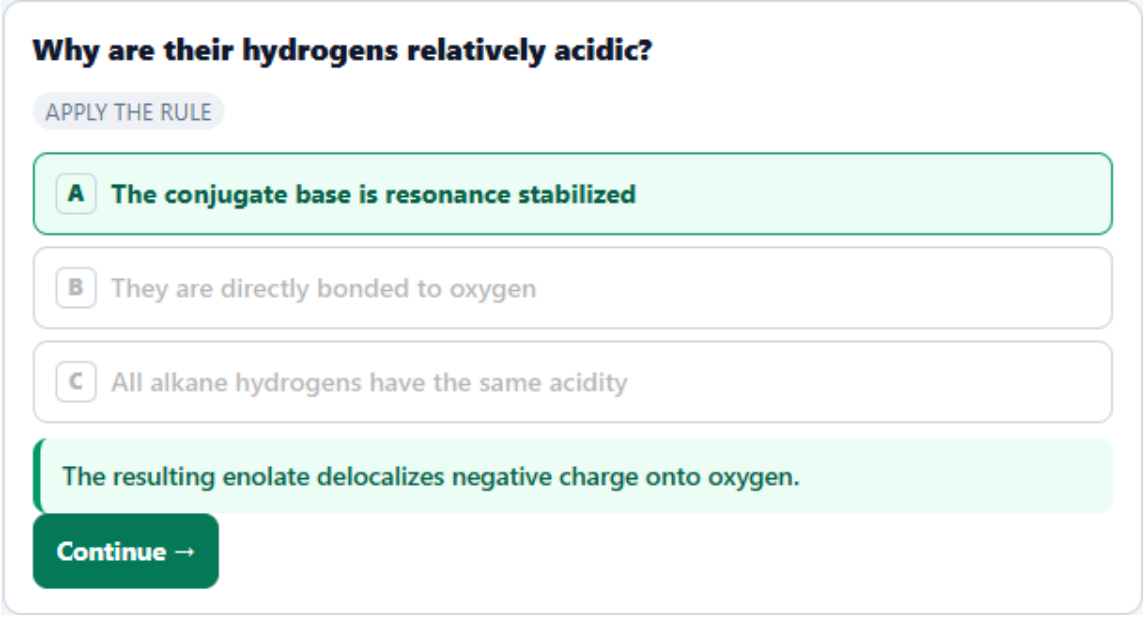
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare the stability of the conjugate base.

Hint 2: Adjacent p orbitals enable delocalization.

Explanation: The resulting enolate delocalizes negative charge onto oxygen.

Screenshot: alpha-map-stage-2-feedback



alpha-map:stage-3 | Enolate resonance contributors differ in…

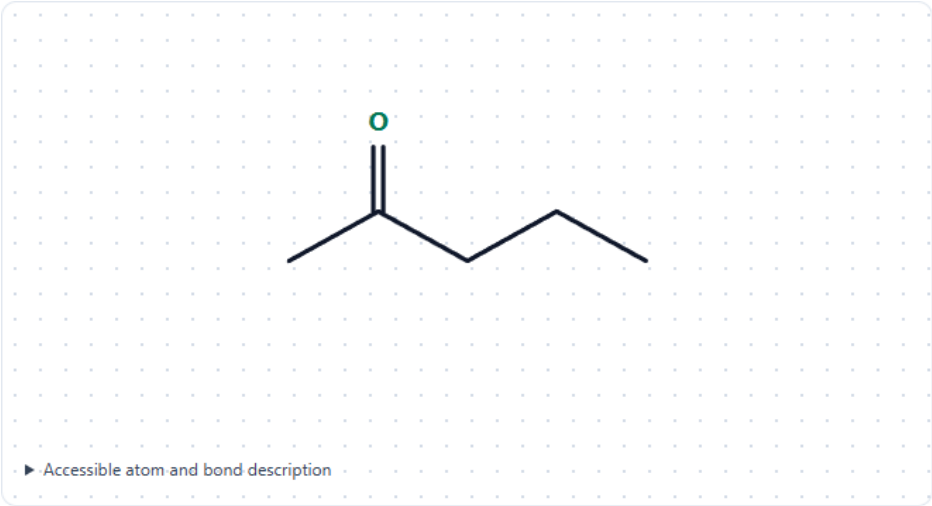
Answer type: choice

Option 0 (A): Electron placement

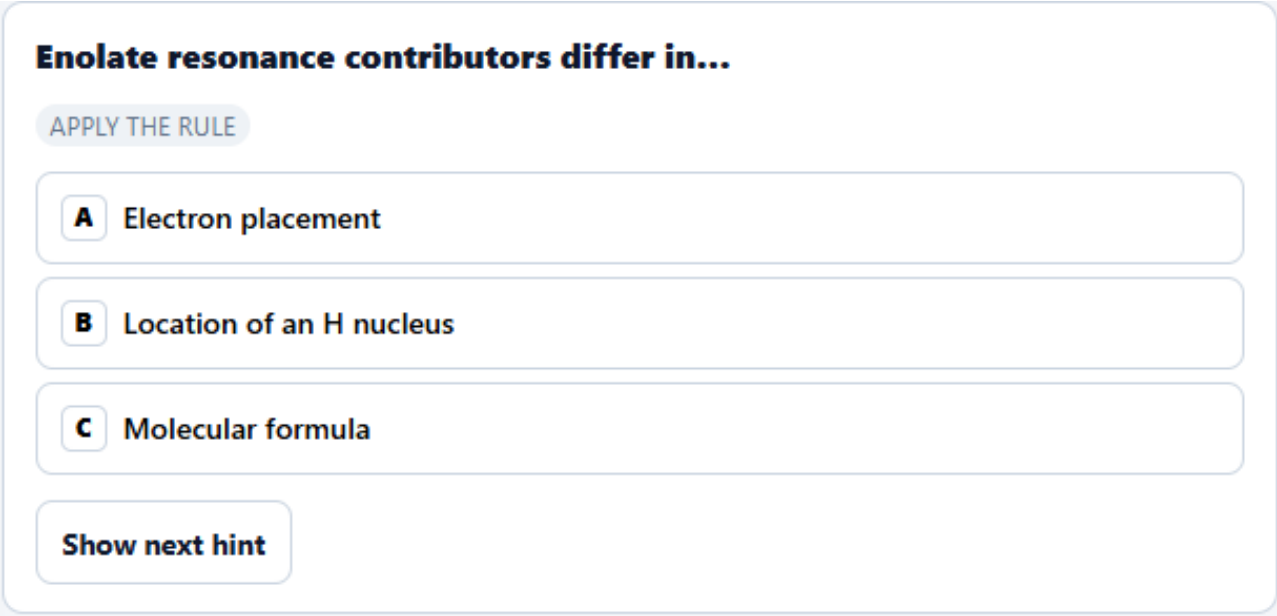
Option 1 (B): Location of an H nucleus

Option 2 (C): Molecular formula

Screenshot: alpha-map-stage-3-molecule



Screenshot: alpha-map-stage-3-question



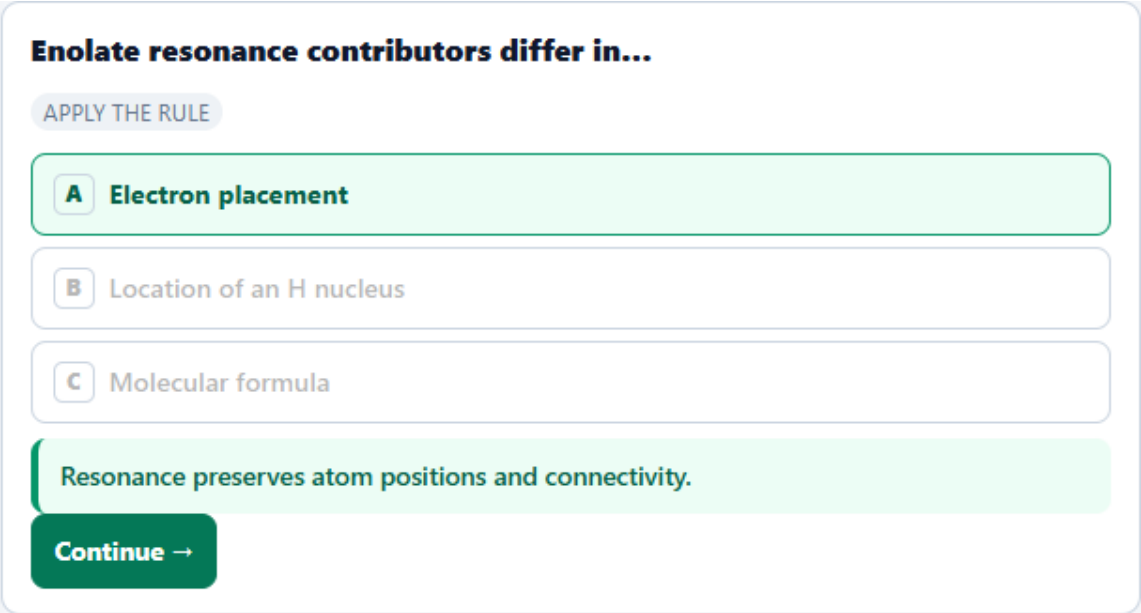
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish resonance from tautomerization.

Hint 2: No atoms move between contributors.

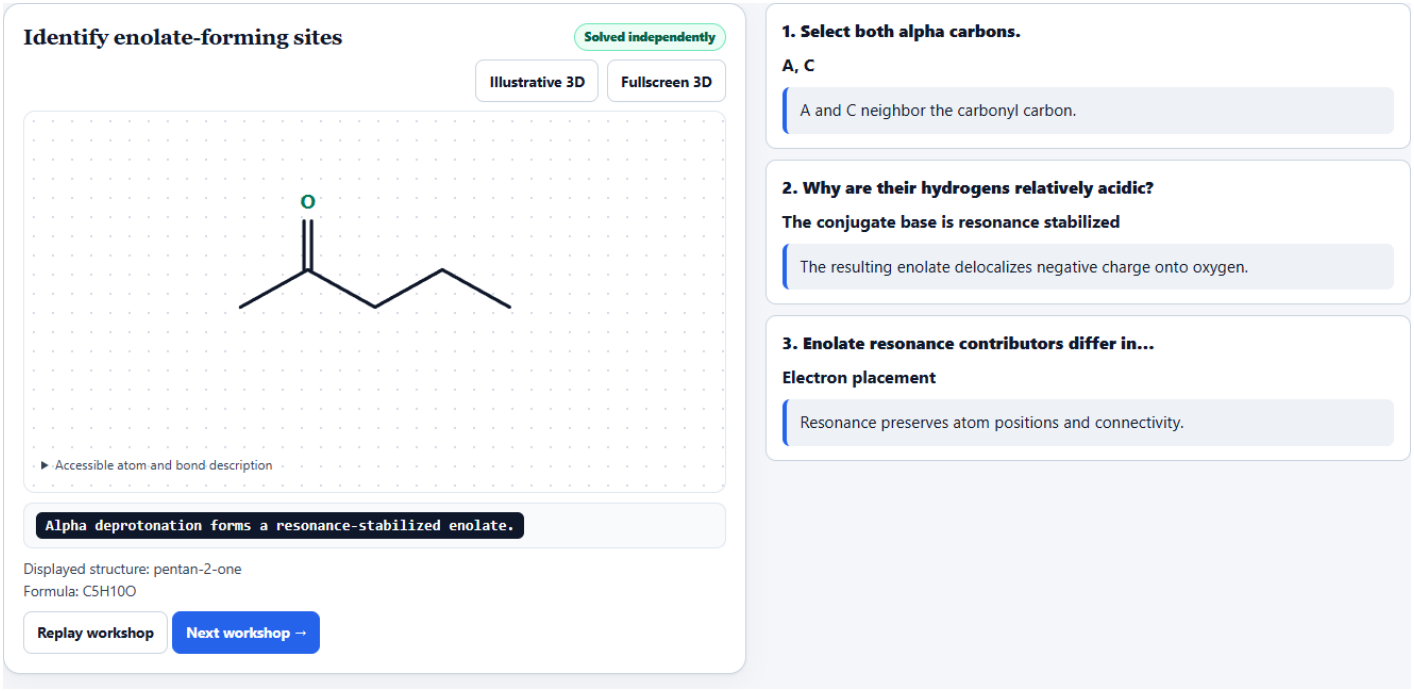
Explanation: Resonance preserves atom positions and connectivity.

Screenshot: alpha-map-stage-3-feedback



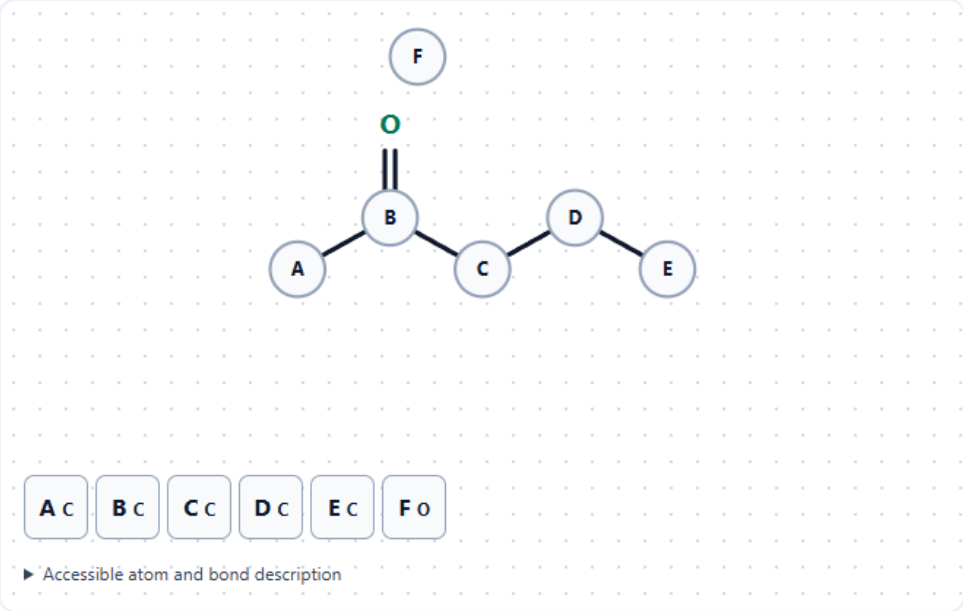
Completion summary: Alpha deprotonation forms a resonance-stabilized enolate.

Screenshot: alpha-map-complete



enolate-control | Choose kinetic or equilibrium control

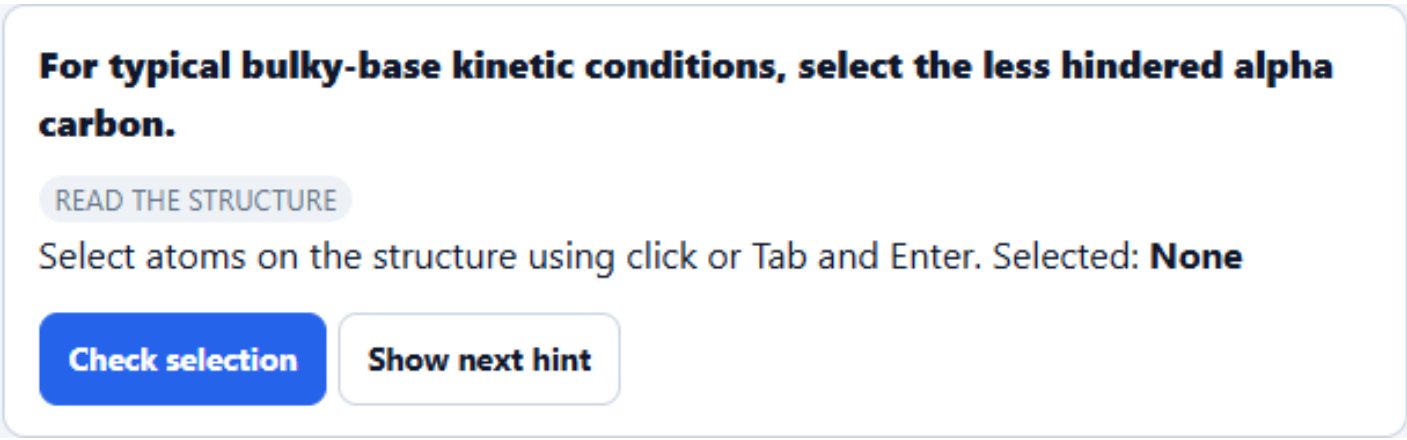
Screenshot: enolate-control-stage-1-molecule



enolate-control:stage-1 | For typical bulky-base kinetic conditions, select the less hindered alpha carbon.

Answer type: atoms

Screenshot: enolate-control-stage-1-question



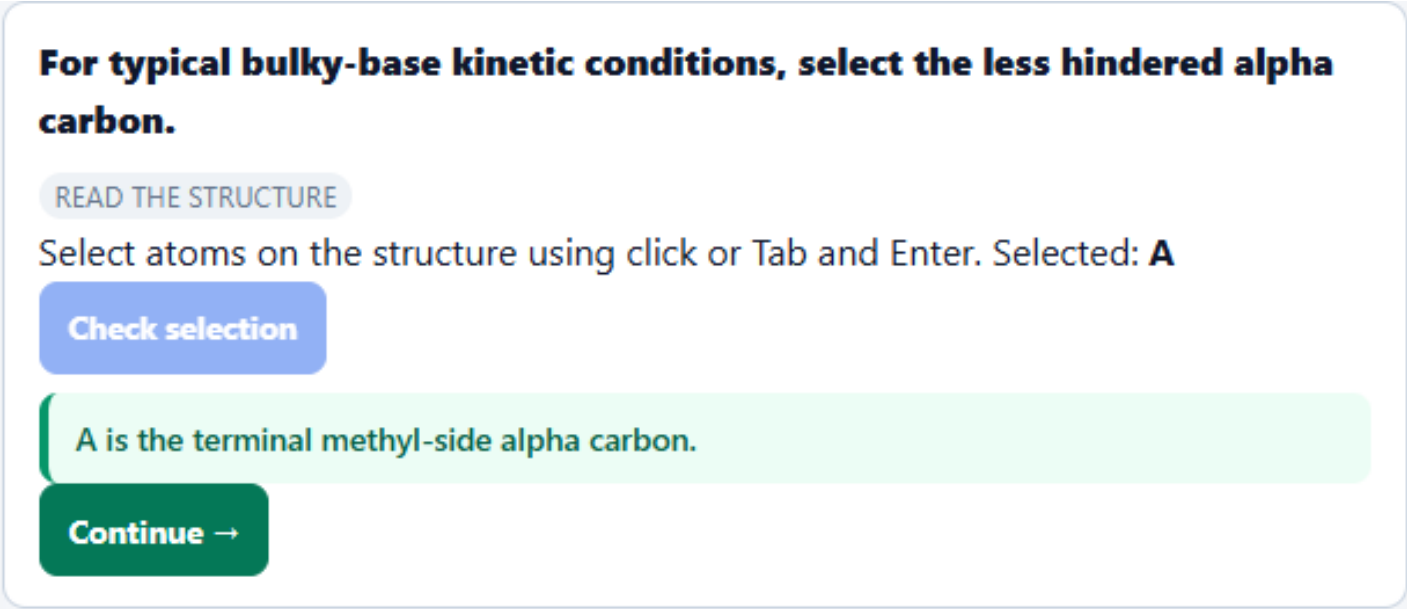
{"authorAnswer": [0], "tolerance": null, "highlightsAfterCorrectAnswer": {"alpha": [0]}}

Hint 1: Compare the two alpha sites.

Hint 2: The terminal methyl side is less substituted.

Explanation: A is the terminal methyl-side alpha carbon.

Screenshot: enolate-control-stage-1-feedback



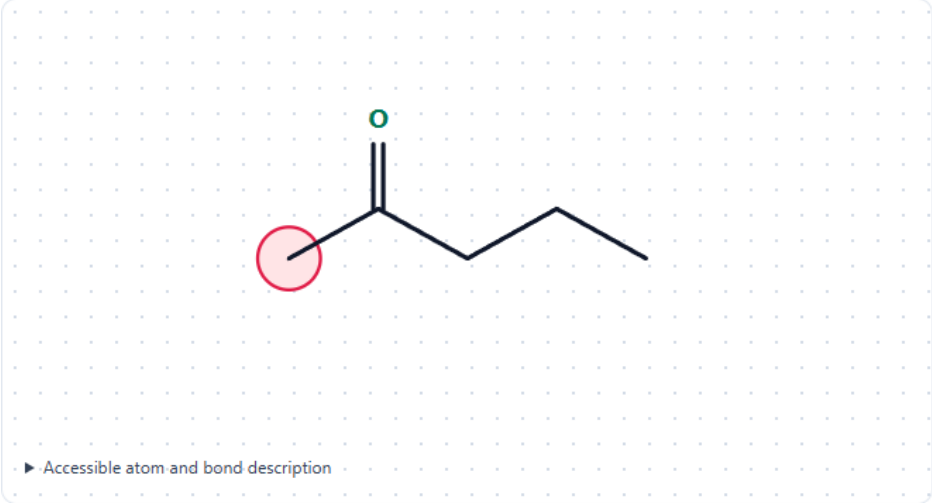
enolate-control:stage-2 | Conditions favoring kinetic enolate formation are usually…

Answer type: choice

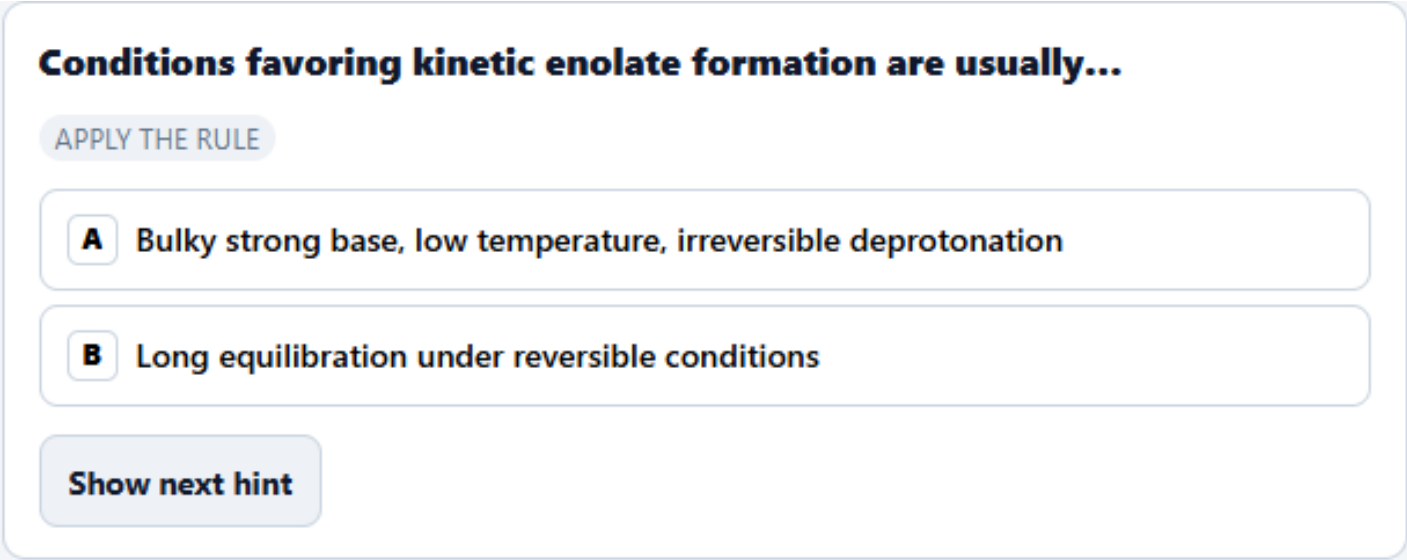
Option 0 (A): Bulky strong base, low temperature, irreversible deprotonation

Option 1 (B): Long equilibration under reversible conditions

Screenshot: enolate-control-stage-2-molecule



Screenshot: enolate-control-stage-2-question



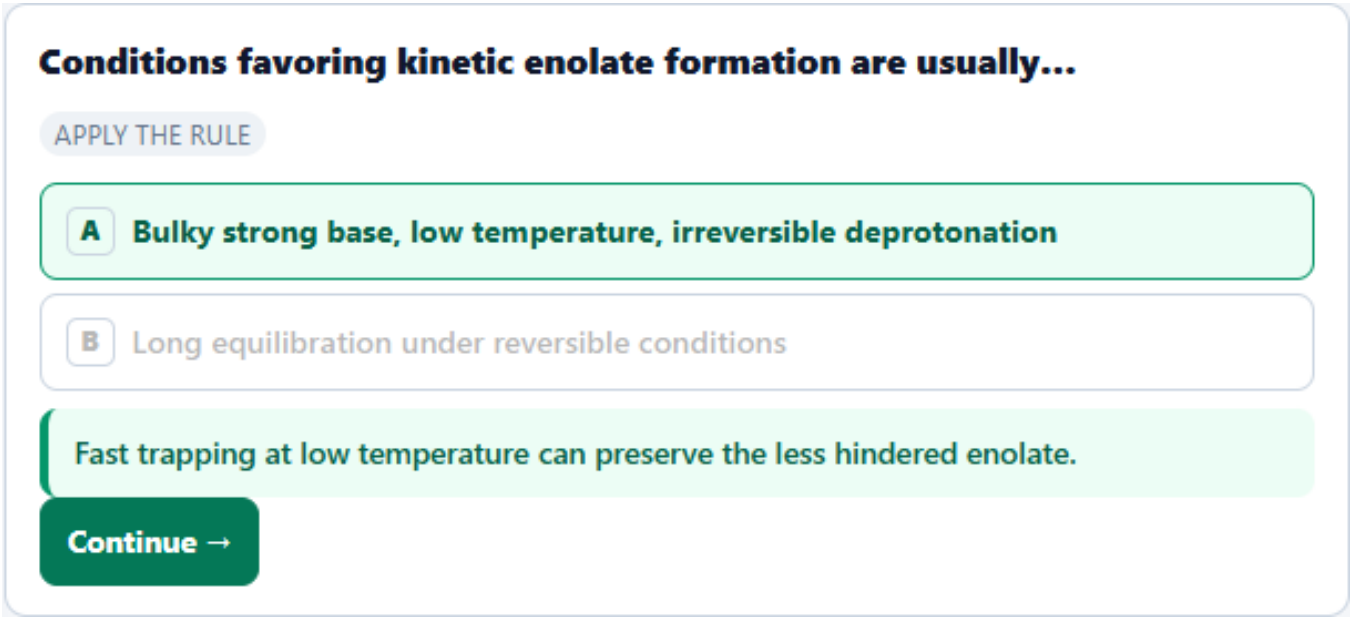
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Kinetic control tracks the lower activation barrier.

Hint 2: Avoid allowing products to equilibrate.

Explanation: Fast trapping at low temperature can preserve the less hindered enolate.

Screenshot: enolate-control-stage-2-feedback



enolate-control:stage-3 | Neutral keto and enol forms are...

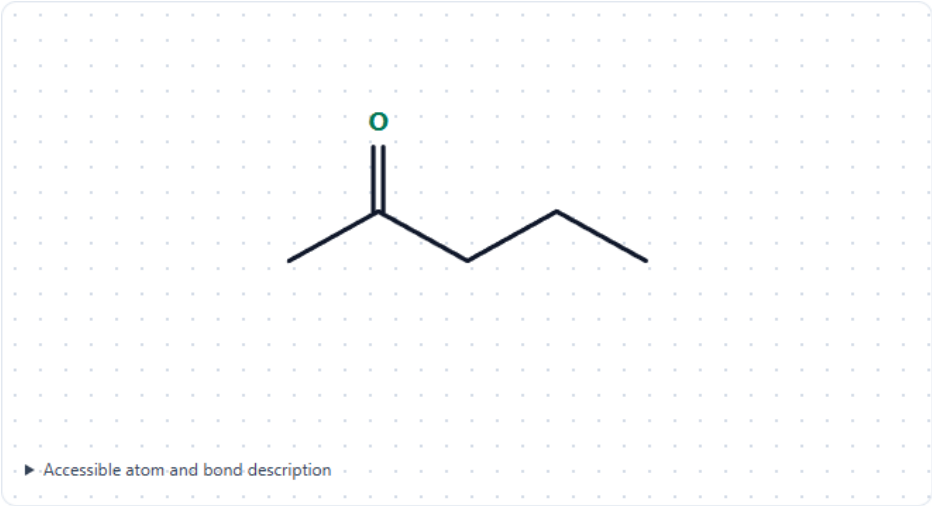
Answer type: choice

Option 0 (A): Tautomers

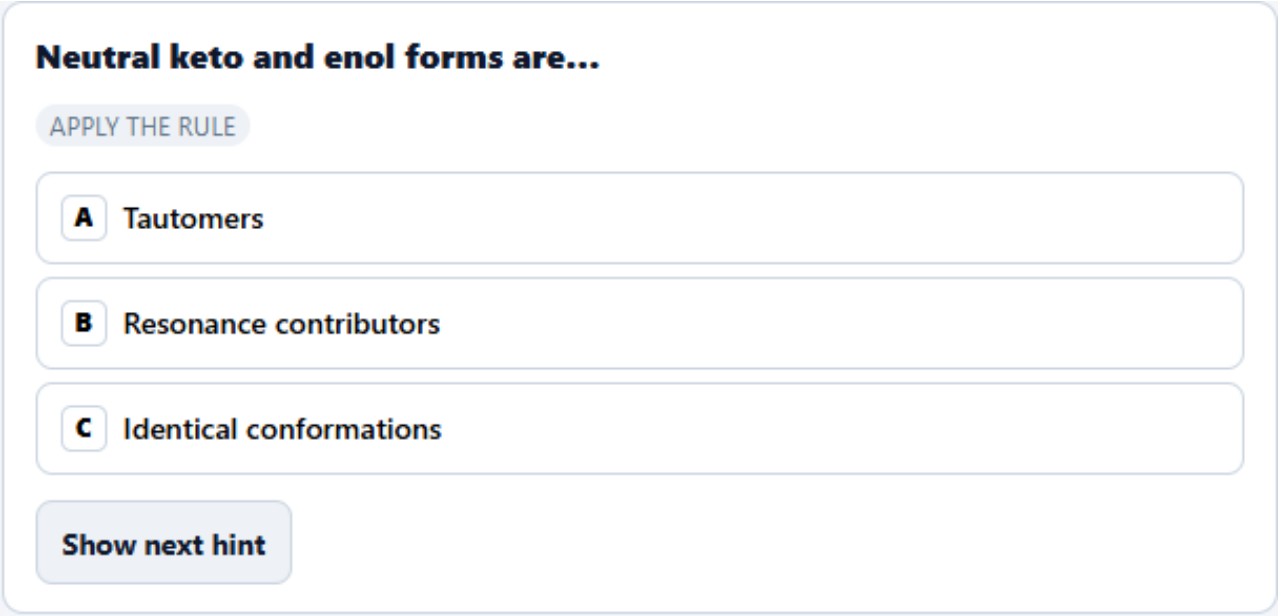
Option 1 (B): Resonance contributors

Option 2 (C): Identical conformations

Screenshot: enolate-control-stage-3-molecule



Screenshot: enolate-control-stage-3-question



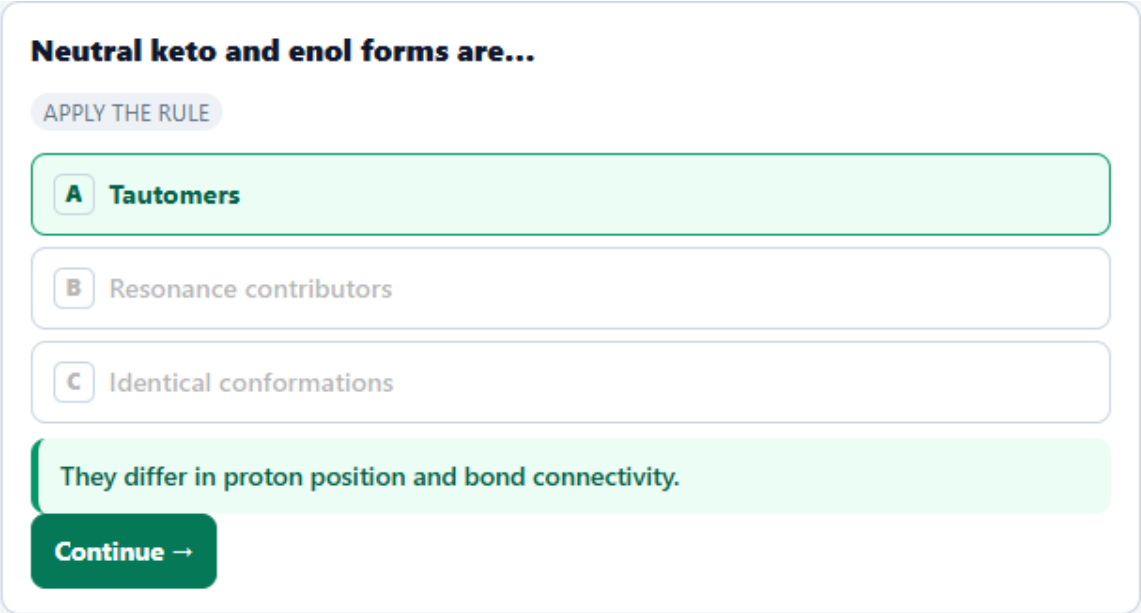
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Check whether an H nucleus moves.

Hint 2: Resonance cannot move nuclei.

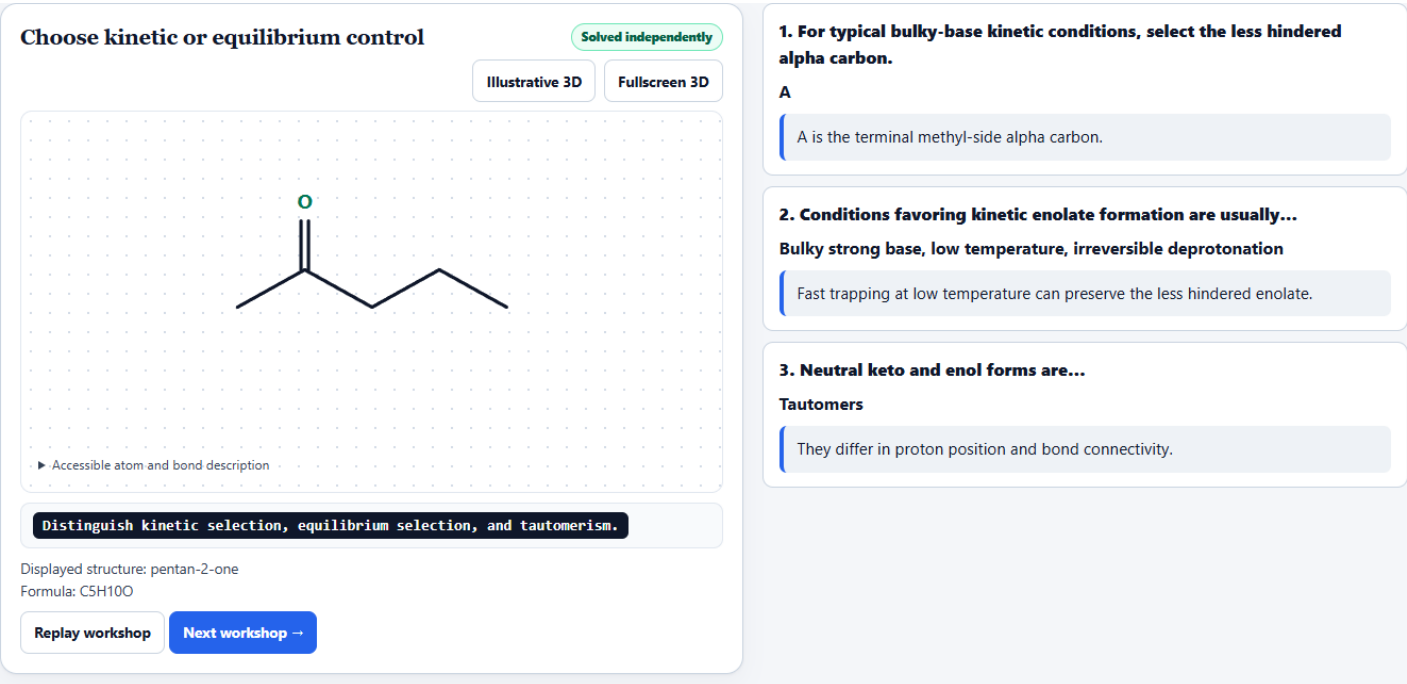
Explanation: They differ in proton position and bond connectivity.

Screenshot: enolate-control-stage-3-feedback



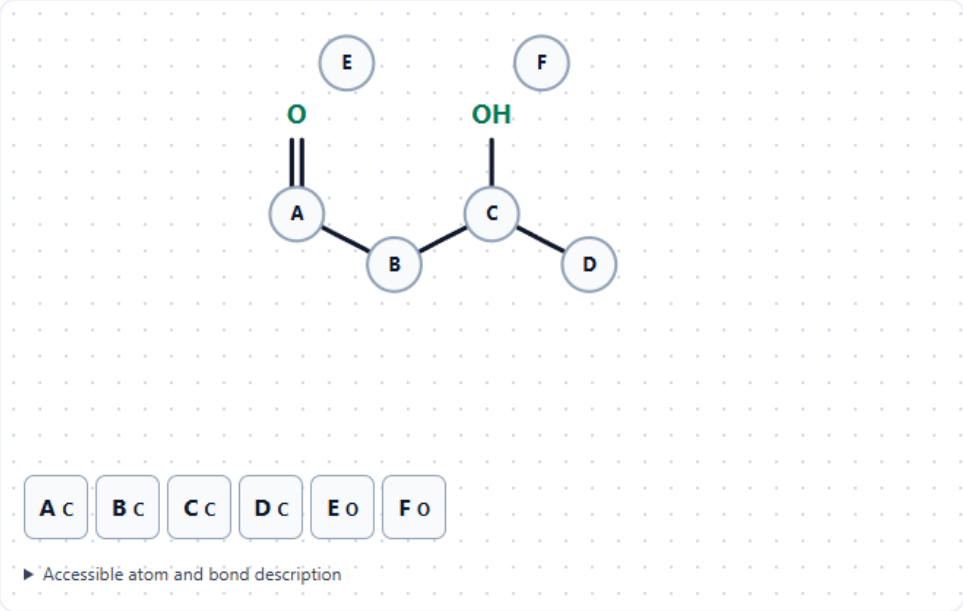
Completion summary: Distinguish kinetic selection, equilibrium selection, and tautomerism.

Screenshot: enolate-control-complete



aldol-map | Trace an aldol product

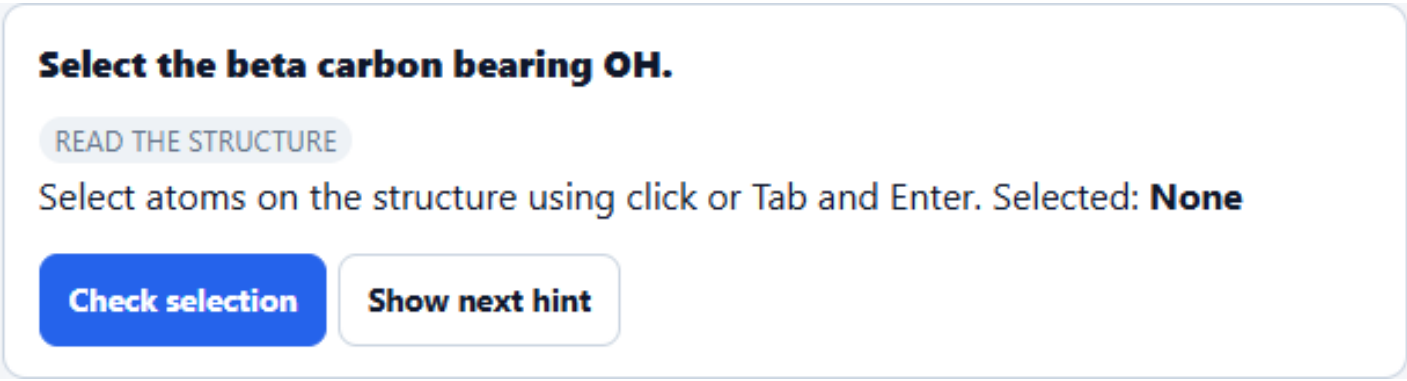
Screenshot: aldol-map-stage-1-molecule



aldol-map:stage-1 | Select the beta carbon bearing OH.

Answer type: atoms

Screenshot: aldol-map-stage-1-question



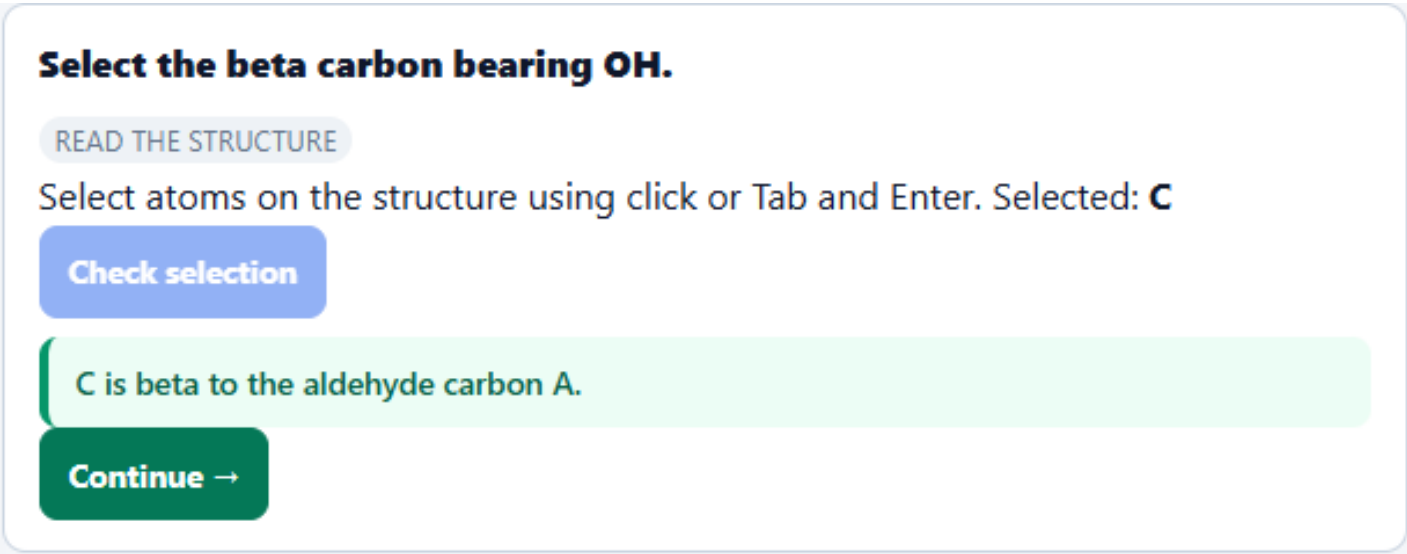
{"authorAnswer": [2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 4], "alpha": [1], "sub": [2, 5]}}

Hint 1: Start at carbonyl carbon A.

Hint 2: B is alpha; the next carbon is beta.

Explanation: C is beta to the aldehyde carbon A.

Screenshot: aldol-map-stage-1-feedback



aldol-map:stage-2 | The displayed beta-hydroxy product is formed by...

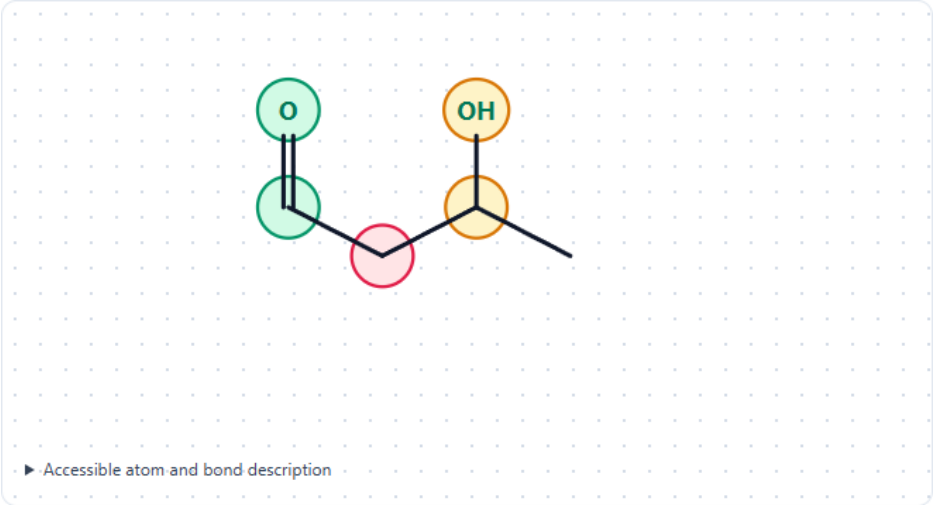
Answer type: choice

Option 0 (A): Aldol addition

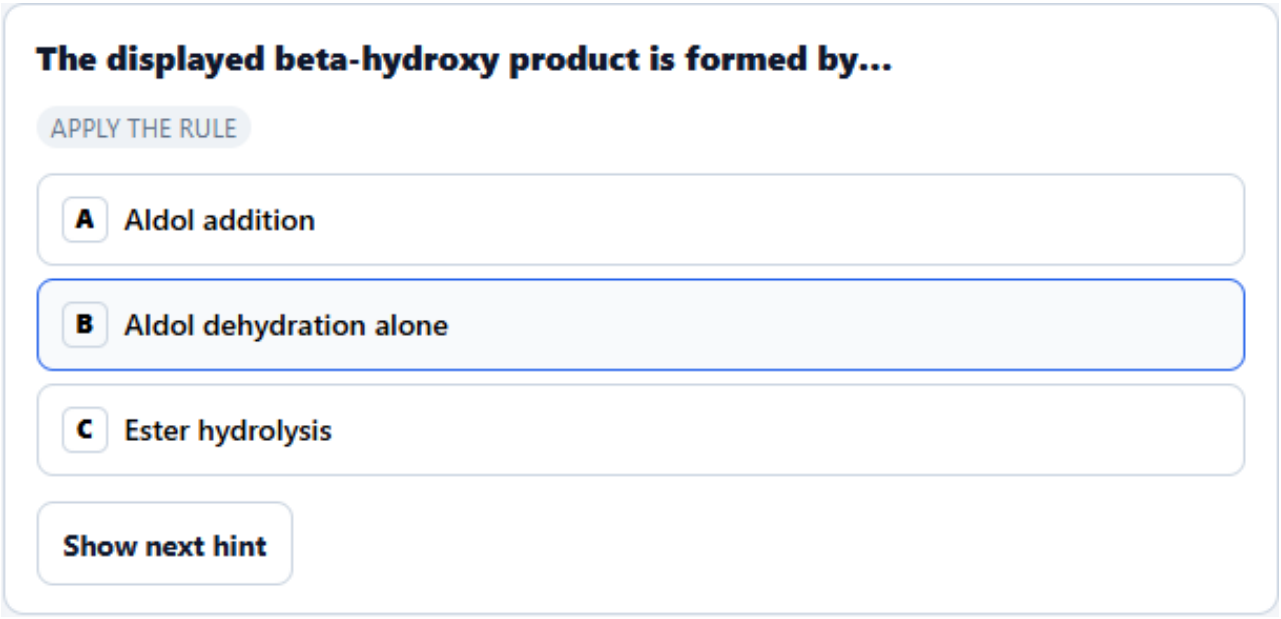
Option 1 (B): Aldol dehydration alone

Option 2 (C): Ester hydrolysis

Screenshot: aldol-map-stage-2-molecule



Screenshot: aldol-map-stage-2-question



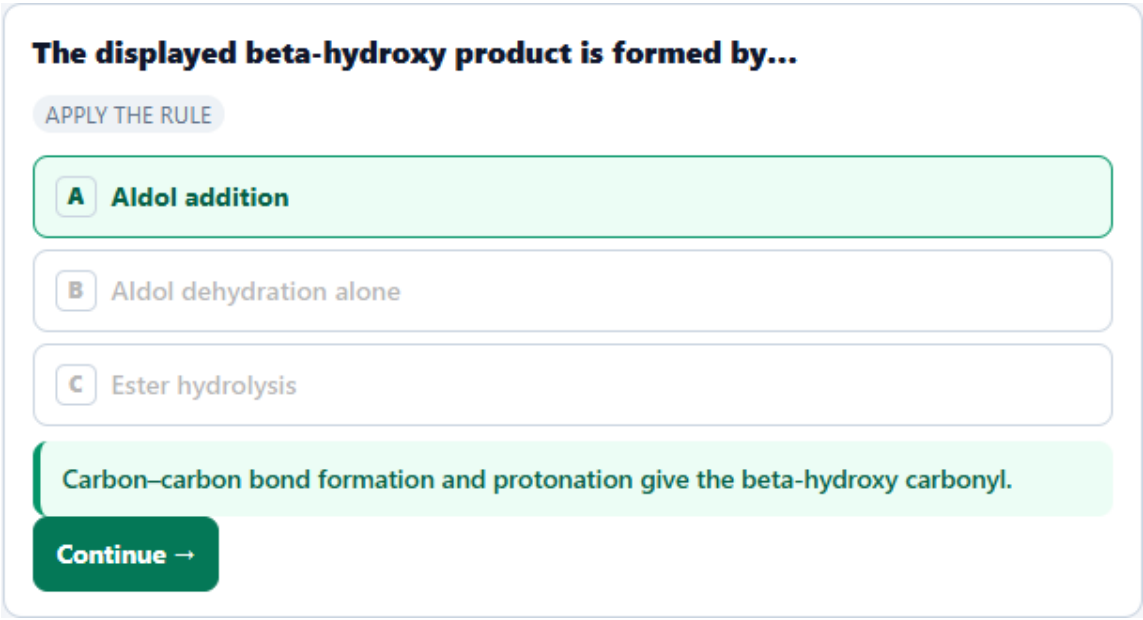
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: The OH is still present.

Hint 2: Water loss has not yet generated a conjugated alkene.

Explanation: Carbon–carbon bond formation and protonation give the beta-hydroxy carbonyl.

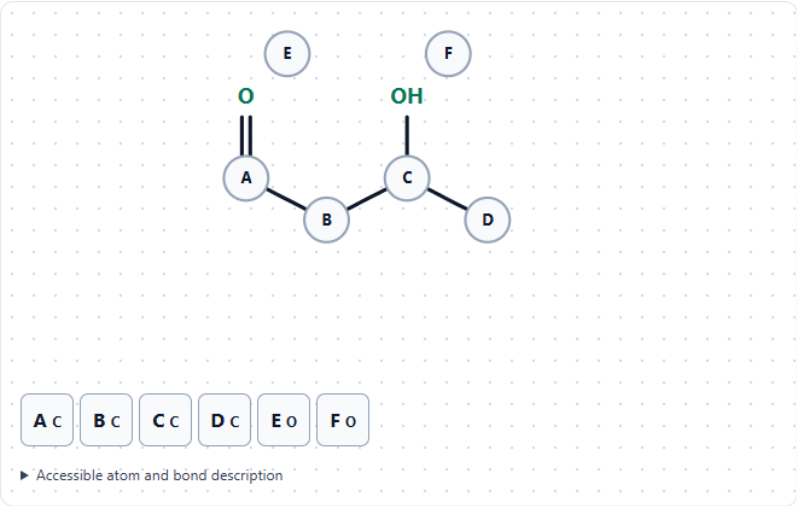
Screenshot: aldol-map-stage-2-feedback



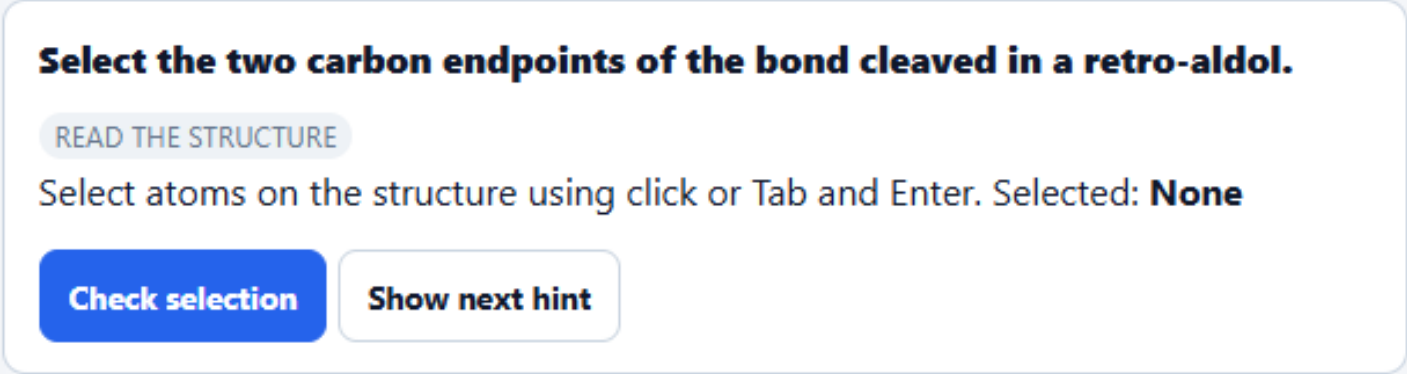
aldol-map:stage-3 | Select the two carbon endpoints of the bond cleaved in a retro-aldol.

Answer type: atoms

Screenshot: aldol-map-stage-3-molecule



Screenshot: aldol-map-stage-3-question



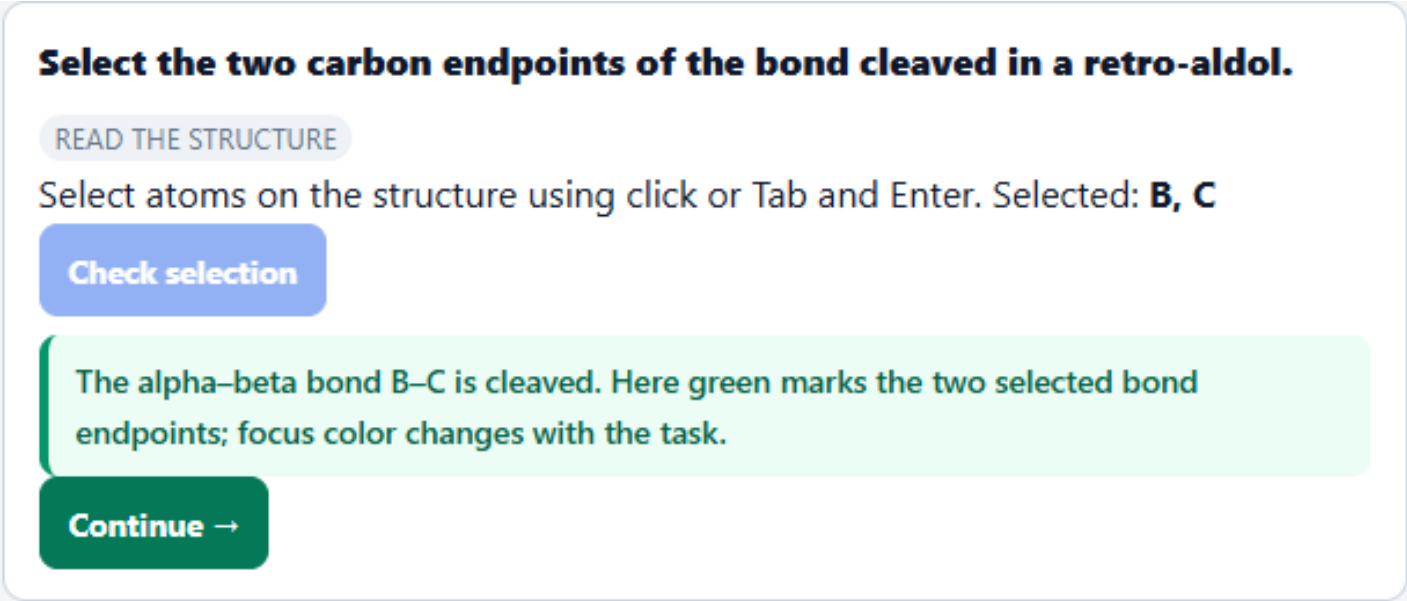
{"authorAnswer": [1, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Reverse the new carbon–carbon bond formation.

Hint 2: Connect the alpha carbon to the OH-bearing beta carbon.

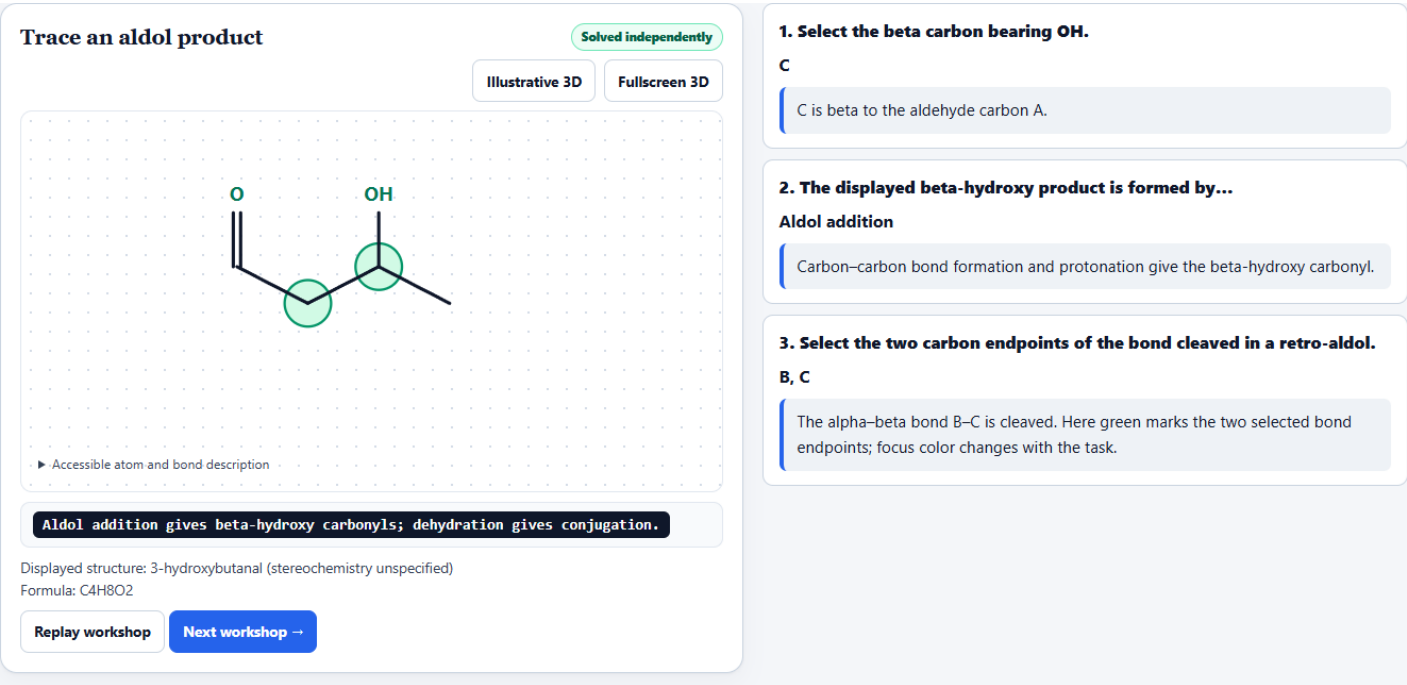
Explanation: The alpha–beta bond B–C is cleaved. Here green marks the two selected bond endpoints; focus color changes with the task.

Screenshot: aldol-map-stage-3-feedback



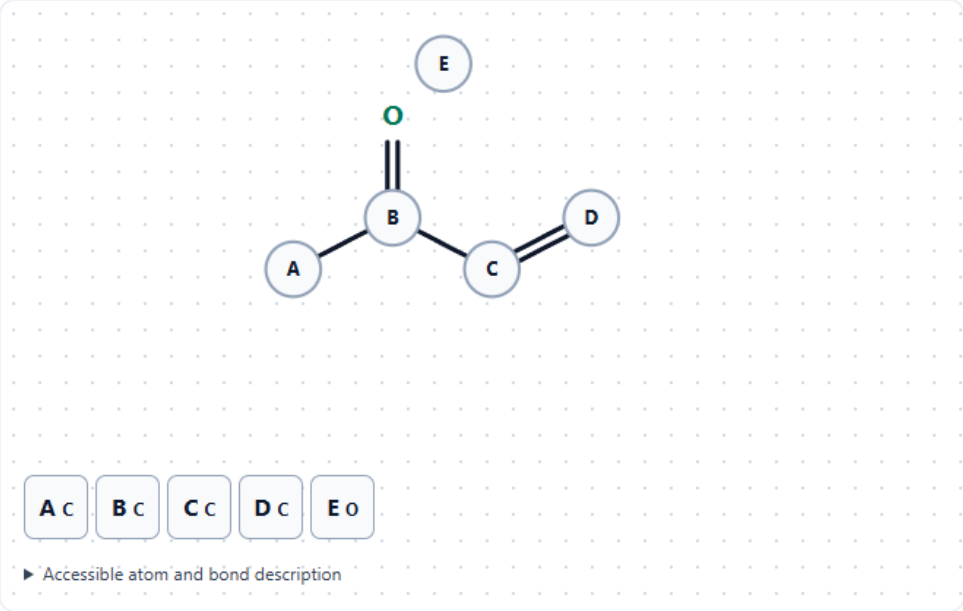
Completion summary: Aldol addition gives beta-hydroxy carbonyls; dehydration gives conjugation.

Screenshot: aldol-map-complete



michael-site | Find a conjugate-addition site

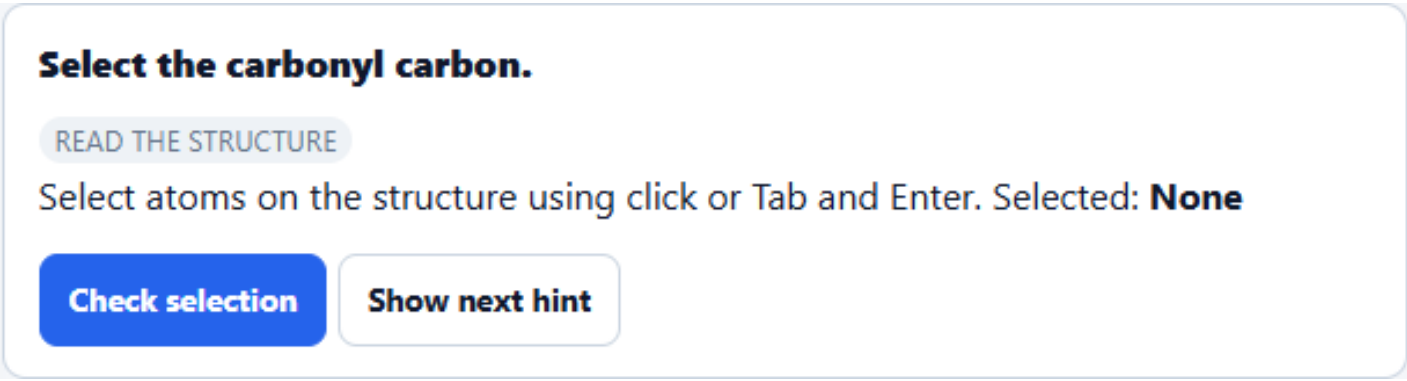
Screenshot: michael-site-stage-1-molecule



michael-site:stage-1 | Select the carbonyl carbon.

Answer type: atoms

Screenshot: michael-site-stage-1-question



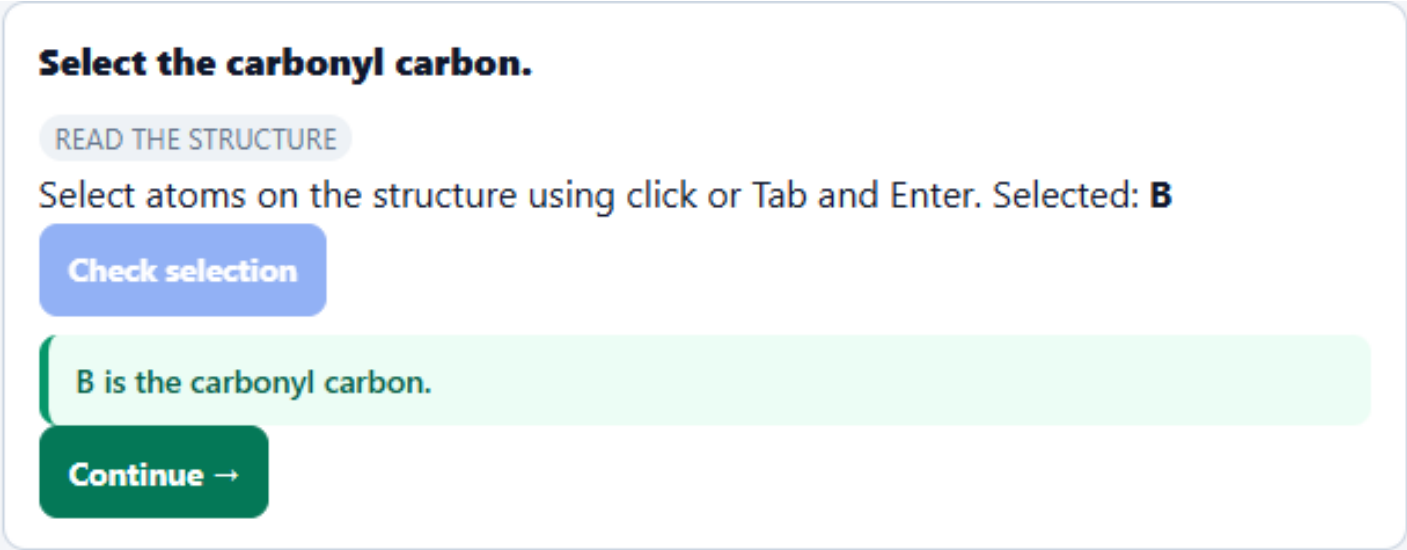
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 4]}}

Hint 1: Find C=O.

Hint 2: Choose the carbon endpoint.

Explanation: B is the carbonyl carbon.

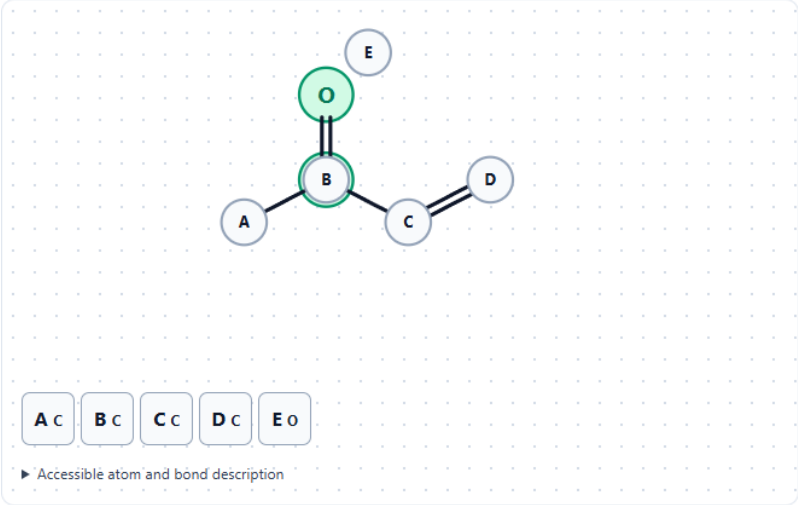
Screenshot: michael-site-stage-1-feedback



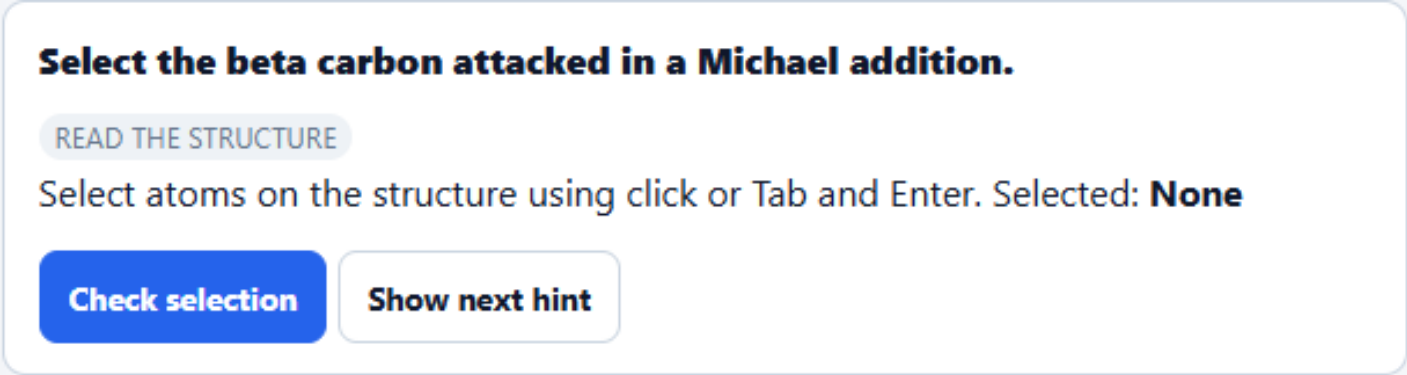
michael-site:stage-2 | Select the beta carbon attacked in a Michael addition.

Answer type: atoms

Screenshot: michael-site-stage-2-molecule



Screenshot: michael-site-stage-2-question



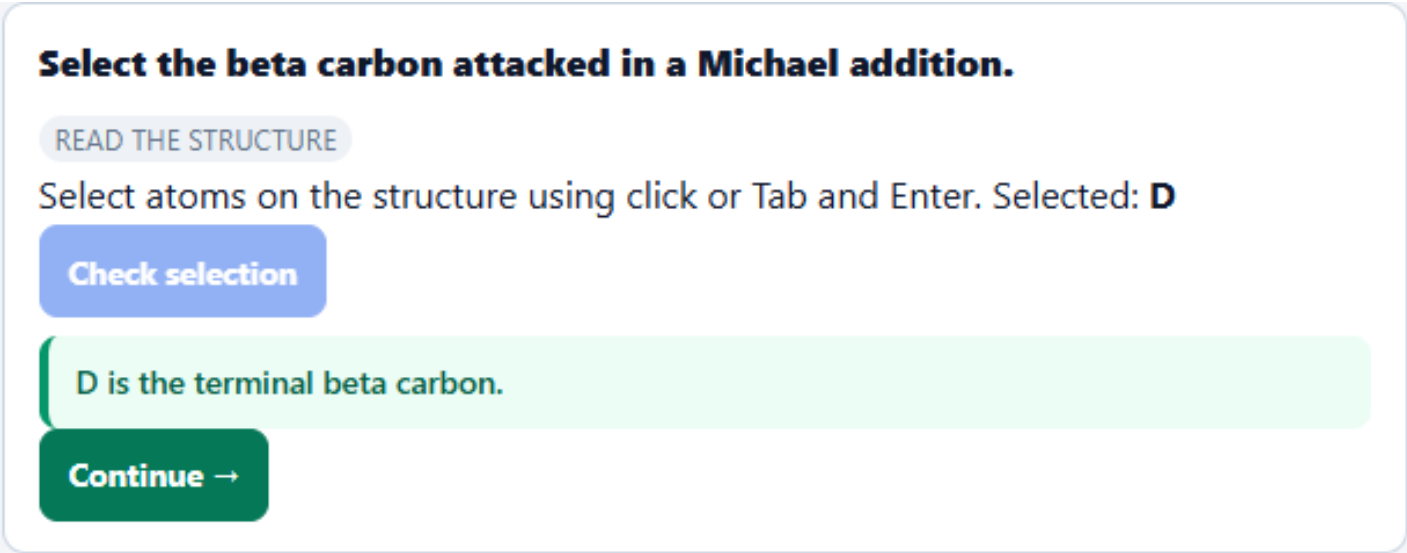
{"authorAnswer": [3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 4], "sub": [3]}}

Hint 1: Move one carbon to alpha, then another to beta.

Hint 2: The alkene extends away from C=O.

Explanation: D is the terminal beta carbon.

Screenshot: michael-site-stage-2-feedback



michael-site:stage-3 | Attack directly at the carbonyl carbon (the C=O site rather than beta) is called…

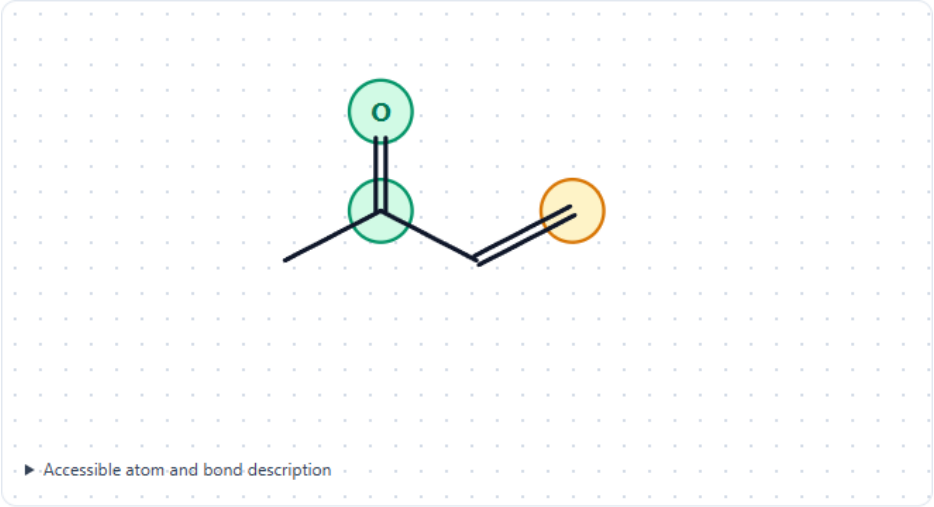
Answer type: choice

Option 0 (A): 1,2-addition

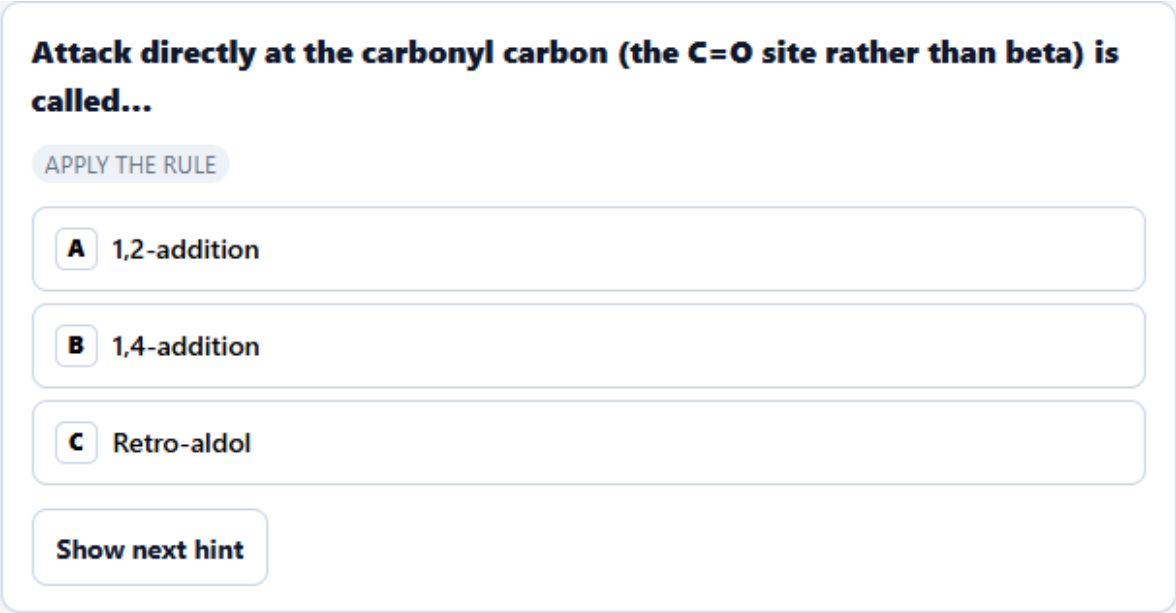
Option 1 (B): 1,4-addition

Option 2 (C): Retro-aldol

Screenshot: michael-site-stage-3-molecule



Screenshot: michael-site-stage-3-question



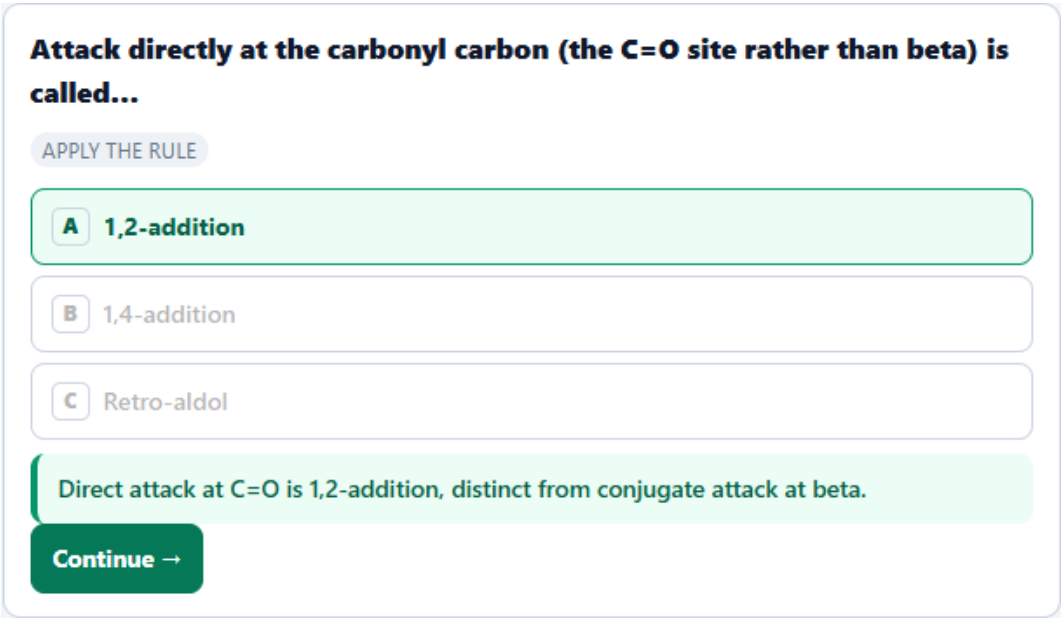
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 4]}}

Hint 1: Track the attacked site.

Hint 2: The oxygen and carbonyl carbon are positions 1 and 2 in this shorthand.

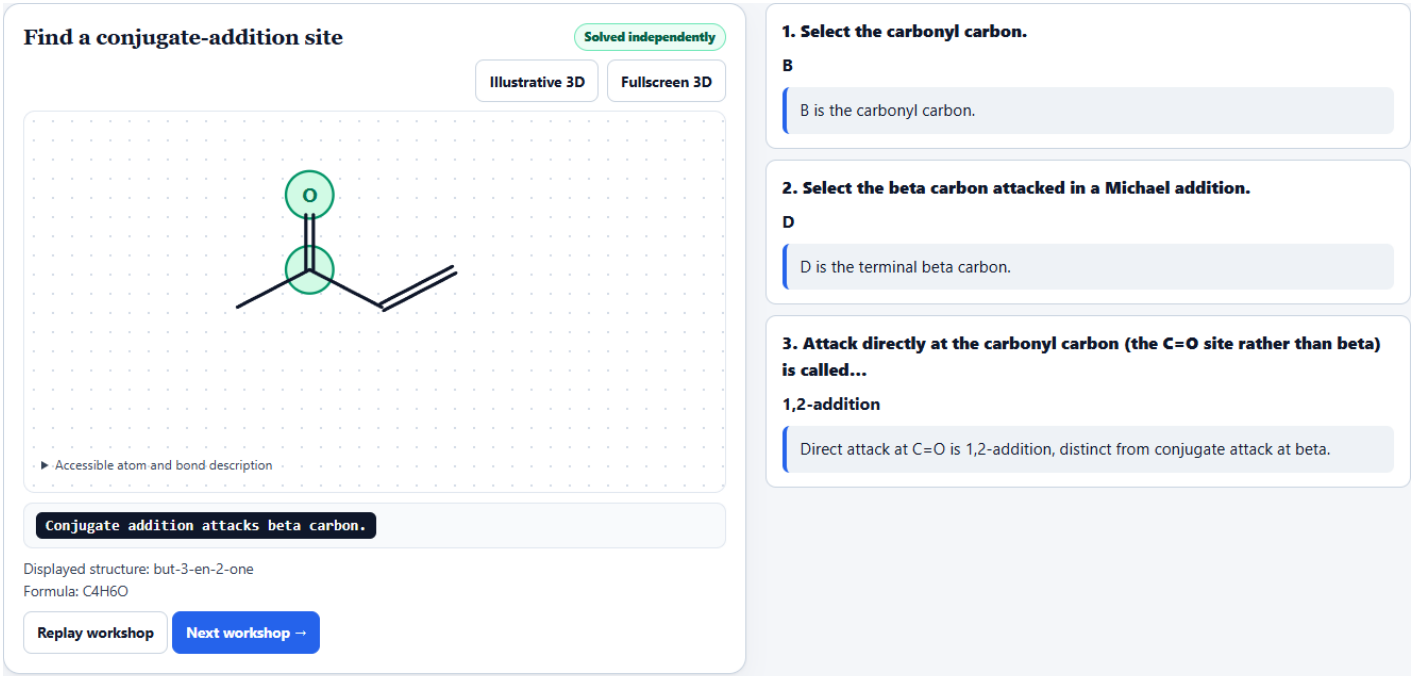
Explanation: Direct attack at C=O is 1,2-addition, distinct from conjugate attack at beta.

Screenshot: michael-site-stage-3-feedback



Completion summary: Conjugate addition attacks beta carbon.

Screenshot: michael-site-complete



Molecular repository: author claims to verify

ketone

Author claims: pentan-2-one | formula C5H10O | canonical composition C5H10O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): O; x=102.00, y=30.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 2]]

ethanal-enolate

Author claims: ethanal enolate (oxygen-charged contributor) | formula C2H3O | canonical composition C2H3O | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=175.00, y=120.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1]]

ethanal-enolate-carbon

Author claims: ethanal enolate (carbon-charged contributor) | formula C2H3O | canonical composition C2H3O | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=120.00; charge=-1; explicit H override=2

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=175.00, y=120.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

vinylalcohol

Author claims: ethenol (vinyl alcohol) | formula C2H4O | canonical composition C2H4O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=165.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1]]

aldol

Author claims: 3-hydroxybutanal (stereochemistry unspecified) | formula C4H8O2 | canonical composition C4H8O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=85.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=115.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=85.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=115.00; charge=0; explicit H override=none

4 (E): O; x=50.00, y=25.00; charge=0; explicit H override=none

5 (F): O; x=166.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [0, 4, 2], [2, 5, 1]]

crotonaldehyde

Author claims: but-2-enal (E/Z unspecified) | formula C4H6O | canonical composition C4H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=85.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=115.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=85.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=115.00; charge=0; explicit H override=none

4 (E): O; x=50.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1], [0, 4, 2]]

methylvinylketone

Author claims: but-3-en-2-one | formula C4H6O | canonical composition C4H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

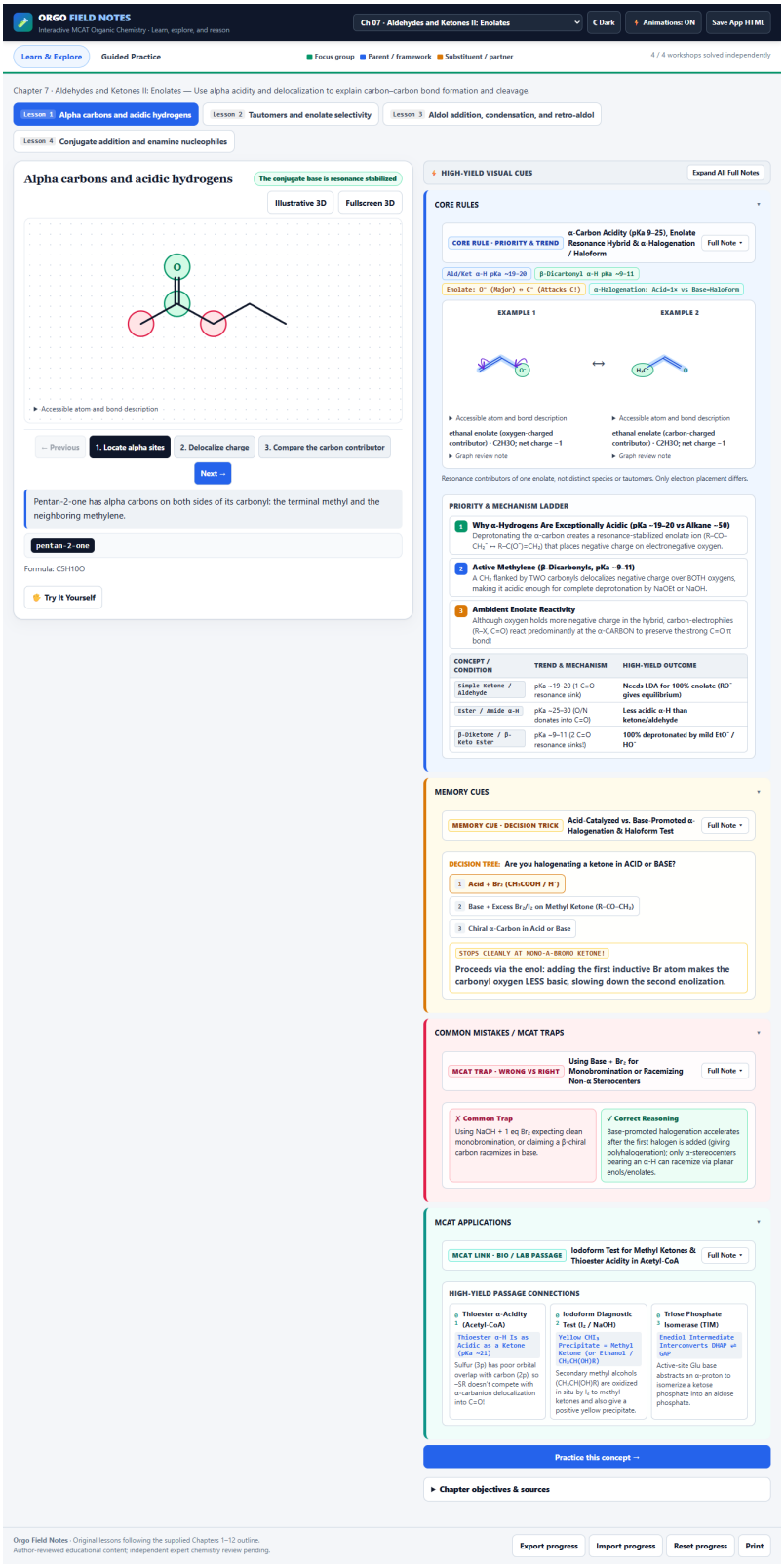
4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [1, 4, 2]]

Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light



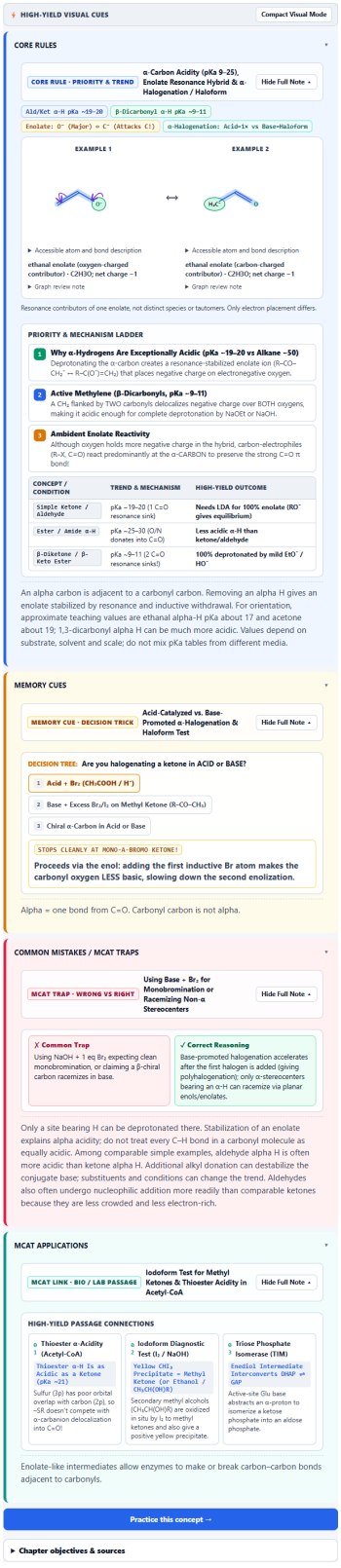
Screenshot: layout-mobile-light



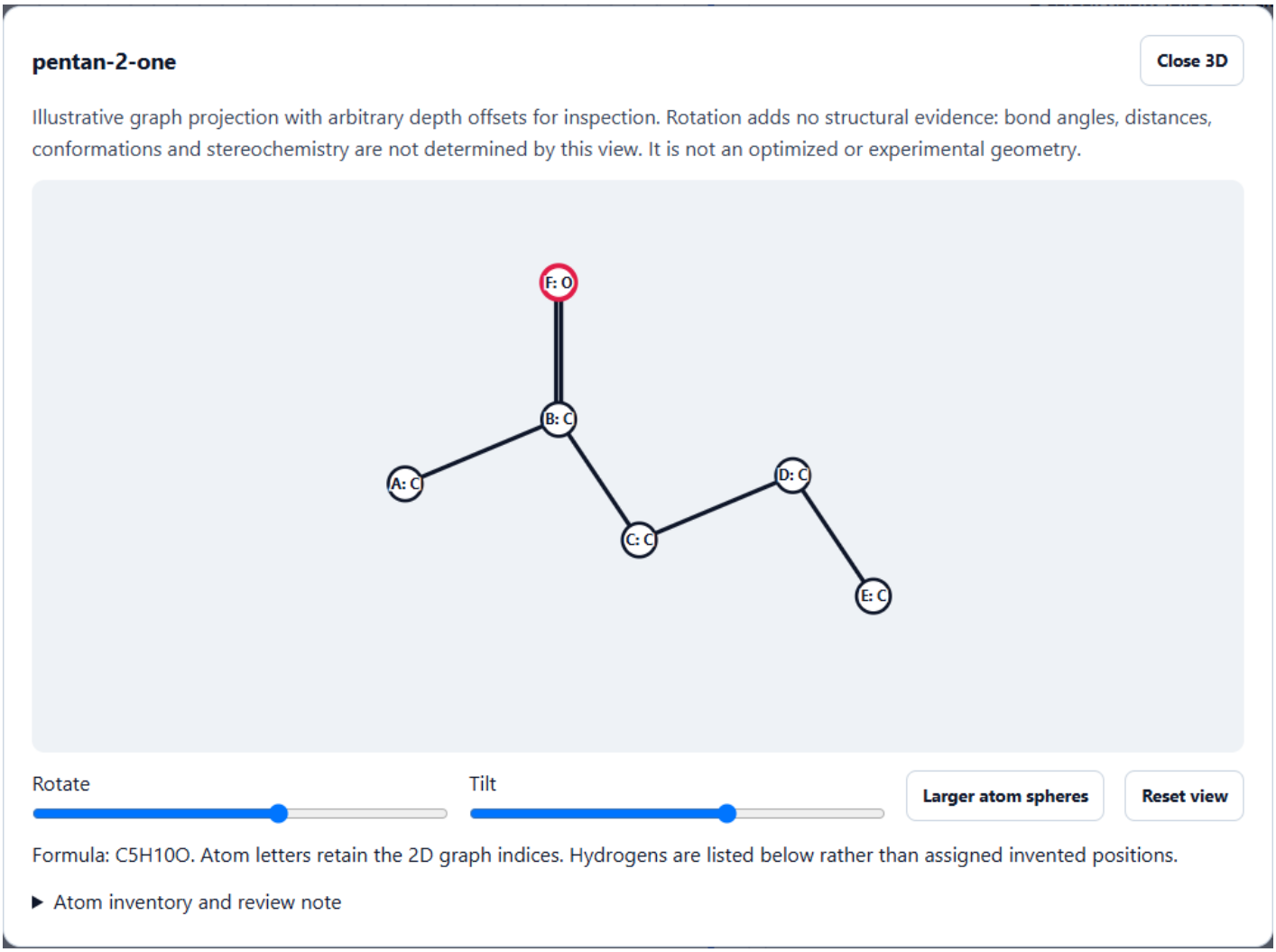
Screenshot: layout-mobile-dark



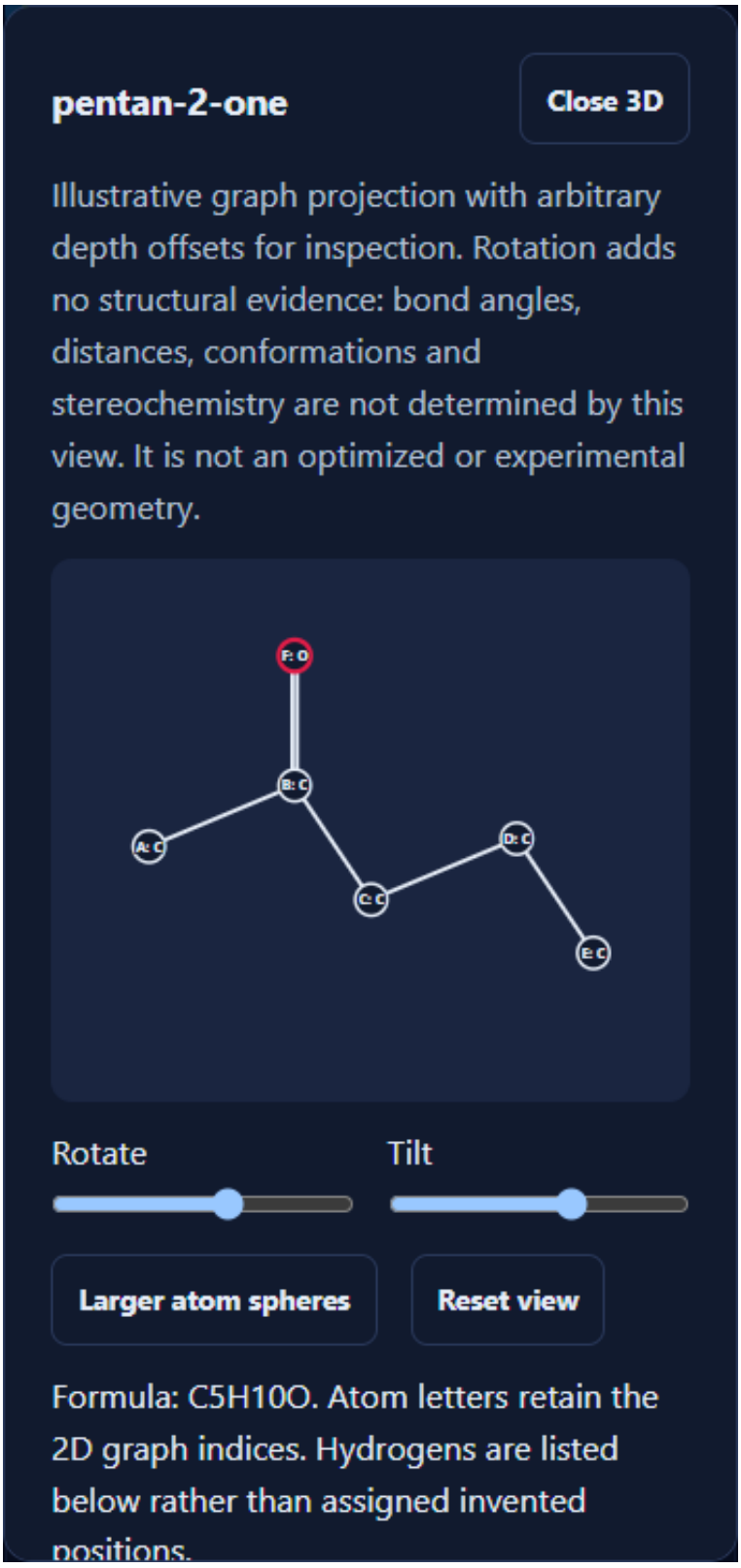
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

63 screenshots and 14 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.